Lesson 1 — Extra Practice

Review: Graphing Linear Equations from a Table of Values

Part A — Tables with Negative and Fractional Inputs

1. Complete each table of values. The x-values are deliberately awkward — keep fractions as fractions and watch every sign.

(a) $y = 4x - 3$

x	-3	$-\frac{1}{2}$	0	$\frac{1}{2}$	2
y					

(b) $y = -2x + 1$

x	$-\frac{5}{2}$	-1	0	$\frac{3}{2}$	4
y					

(c) $y = \frac{1}{2}x + 4$

x	-6	-3	0	1	5
y					

(d) $y = -\frac{2}{3}x - 1$

x	-6	-3	0	$\frac{3}{2}$	6
y					

2. Each table below has **gaps in both rows**. Use the given equation to fill every empty cell. Some gaps need you to work forwards, others backwards.

(a) $y = 5x - 2$

x	-2		0		3
y	-12	-7		8	

(b) $y = -3x + 4$

x	-3	-1		2	
y	13		4		-11

Part B — Finding the Rule and Testing for Linearity

3. Each table below comes from a linear equation. Find the rule and write it in the form $y = mx + b$ (m is the rate: the change in y for each step of 1 in x; b is the y-value at $x = 0$). In part (c) the x-values step by 2, not by 1 — divide the change in y by the change in x.

(a)

x	-2	-1	0	1	2
y	-7	-4	-1	2	5

(b)

x	-2	-1	0	1	2
y	9	7	5	3	1

(c)

x	-4	-2	0	2	4
y	-1	0	1	2	3

4. **When first differences mislead.** Priya looks at the table below and says "the y-values go up by 6, then 3, then 9, then 6 — the jumps are not equal, so this relation is *not* linear."

x	-3	-1	0	3	5
y	-11	-5	-2	7	13

(a) Explain the flaw in Priya's reasoning. What did she forget to look at?
(b) For each pair of neighbouring columns, compute $\dfrac{\text{change in } y}{\text{change in } x}$. What do you notice?
(c) Write the equation of the relation, then use it to find y when $x = -8$.

5. Decide whether each table is **linear** or **non-linear**. Justify each answer with a rate, not just a glance. For the linear ones, write the equation.

(a)

x	0	1	2	3
y	1	4	9	16

(b)

x	-1	0	1	2
y	5	2	-1	-4

(c)

x	1	2	3	4
y	2	4	8	16

(d)

x	-2	0	2	4
y	-3	1	5	9

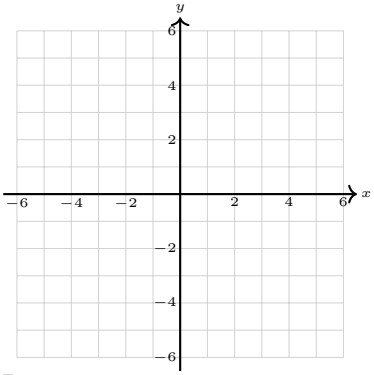
6. Riley sees the table $x : 1, 2, 3, 4$ with $y : 2, 4, 8, 16$ and says "there is a clear pattern, so the relation must be linear." Write a short reply explaining why *having a pattern* is not the same as *being linear*, and state exactly what a table must show to be linear.

Part C — Plotting from a Table, and Reading a Graph

7. For each equation: complete the table, plot the five points, and draw the line right across the grid. Then describe the pattern in one sentence — how far the line climbs or falls each time x increases by 1, and where it crosses the y-axis.

(a) $y = \dfrac{1}{2}x - 1$

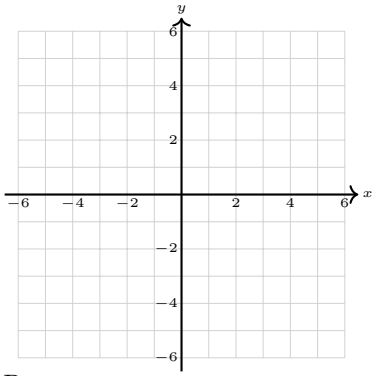
x	-4	-2	0	2	4
y					



Pattern: _____

(b) $y = -2x + 3$

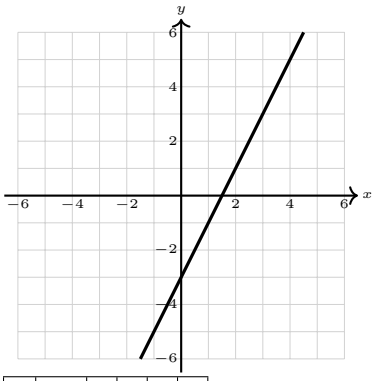
x	-1	0	1	2	3
y					



Pattern: _____

8. Each line below is already drawn. Read the graph to complete the table, then write the equation of the line.

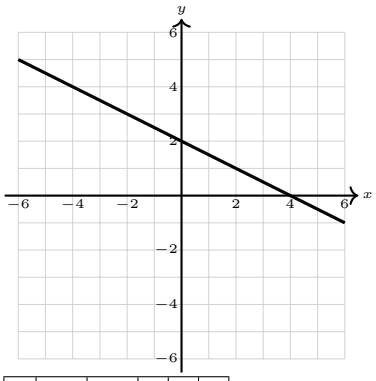
(a)



x	-1	0	1	2	3
y					

Equation: _____

(b)



x	-4	-2	0	2	4
y					

Equation: _____

Part D — Working Backwards, Errors, and Applications

9. Work backwards. For each equation, determine the x-value that produces the given y-value. Show the substitution and solve.

(a) $y = 4x - 5$; find x when $y = 31$.

(b) $y = -3x + 7$; find x when $y = 31$.

(c) $y = \frac{2}{3}x + 5$; find x when $y = 31$.

(d) $y = 6x + 1$; find x when $y = -29$.

10. A linear relation contains the points $(2, 7)$ and $(5, 16)$. Without graphing, determine the y-value when $x = 10$. Explain how you found the rate of change.

11. **Error analysis.** For $y = 3x - 4$, Jordan wanted the point where $x = 5$, but wrote $5 = 3y - 4$ and got $y = 3$. Explain exactly what Jordan swapped, then give the correct ordered pair.

12. Determine whether each point lies on the given line. Substitute carefully — the coordinates are fractions.

(a) Does $\left(-\frac{1}{2}, 4\right)$ lie on $y = -6x + 1$?

(b) Does $\left(\frac{2}{3}, 0\right)$ lie on $y = 3x - 1$?

(c) Does $(-4, 13)$ lie on $y = -\frac{3}{2}x + 7$?

13. A print shop charges a fixed setup fee plus a cost per poster. The table shows the total cost C, in dollars, for n posters.

n	0	2	5		12
C	8		23	35	

(a) Fill in the three missing entries.
(b) State the cost per poster and the setup fee, and say how you can see each one in the table.
(c) Write the equation relating C and n.
(d) How many posters can be printed for exactly \$59?

14. A tank is draining at a steady rate. The readings below were **not** taken at equal time intervals.

t (min)	0	3	5	10	12
V (L)	90	75	65	40	30

(a) Show that the tank drains at a constant rate, and state that rate with its units.
(b) Write the equation relating V and t.
(c) How much water is left after 8 minutes?
(d) At what time is the tank empty?

15. **Stretch.** Two linear relations begin as shown.

Relation A

x	0	1	2	3
y	1	4	7	10

Relation B

x	0	1	2	3
y	11	9	7	5

(a) Write the equation of each relation. (b) Find the one x-value at which the two relations give the same y-value, and state that ordered pair. (c) For x-values larger than that, which relation gives the greater y? Explain using the two rates.

16. *(Multiple Choice)* A linear relation contains $(-2, 11)$ and $(4, -7)$. Which point also lies on this relation?
A. $(0, -3)$ B. $(1, 2)$ C. $(3, -2)$ D. $(5, 10)$

17. *(Numerical Response)* The point $(k, -17)$ lies on the line $y = -4x + 3$. The value of k is _____.

Answer Key. 1. (a) $-15, -5, -3, -1, 5$ (b) $6, 3, 1, -2, -7$ (c) $1, \frac{5}{2}, 4, \frac{9}{2}, \frac{13}{2}$ (d) $3, 1, -1, -2, -5$. **2.** (a) missing x: -1 and 2; missing y: -2 (at $x = 0$) and 13 (at $x = 3$). (b) missing y: 7 (at $x = -1$), -2 (at $x = 2$); missing x: 0 (where $y = 4$) and 5 (where $y = -11$). **3.** (a) $y = 3x - 1$ (b) $y = -2x + 5$ (c) $y = \frac{1}{2}x + 1$. **4.** (a) She compared changes in y without checking the matching changes in x; the x-steps here are $2, 1, 3, 2$, not all equal. (b) $\frac{6}{2} = \frac{3}{1} = \frac{9}{3} = \frac{6}{2} = 3$ — the rate is a constant 3, so the relation *is* linear. (c) $y = 3x - 2$; at $x = -8$, $y = -26$. **5.** (a) Non-linear: rates $3, 5, 7$. (b) Linear, rate -3: $y = -3x + 2$. (c) Non-linear: rates $2, 4, 8$ (y doubles). (d) Linear, rate $\frac{4}{2} = 2$: $y = 2x + 1$. **6.** A pattern only has to be predictable; linear means the *rate* $\frac{\text{change in } y}{\text{change in } x}$ is the same between every pair of columns. Here y doubles $(+2, +4, +8)$, so the rate grows and the graph curves. **7.** (a) $y : -3, -2, -1, 0, 1$; the line rises 1 for every 2 right (up $\frac{1}{2}$ each step of 1) and crosses the y-axis at $(0, -1)$. (b) $y : 5, 3, 1, -1, -3$; the line falls 2 each time x increases by 1 and crosses the y-axis at $(0, 3)$. **8.** (a) $y : -5, -3, -1, 1, 3$; $y = 2x - 3$. (b) $y : 4, 3, 2, 1, 0$; $y = -\frac{1}{2}x + 2$. **9.** (a) $x = 9$ (b) $x = -8$ (c) $x = 39$ (d) $x = -5$. **10.** Rate $= \frac{16-7}{5-2} = 3$, so $y = 3x + 1$; at $x = 10$, $y = 31$. **11.** Jordan substituted 5 for y instead of for x — the given number belongs in the x-slot. Correctly, $y = 3(5) - 4 = 11$, so the point is $(5, 11)$. **12.** (a) Yes: $-6\left(-\frac{1}{2}\right) + 1 = 3 + 1 = 4$. (b) No: $3\left(\frac{2}{3}\right) - 1 = 1 \neq 0$. (c) Yes: $-\frac{3}{2}(-4) + 7 = 6 + 7 = 13$. **13.** (a) $C = 14$ at $n = 2$; $n = 9$ where $C = 35$; $C = 44$ at $n = 12$. (b) \$3 per poster (the cost rises \$15 over 5 posters, or \$6 over 2); setup fee \$8 — the cost when $n = 0$. (c) $C = 3n + 8$. (d) $3n + 8 = 59$ gives $n = 17$ posters. **14.** (a) Every interval gives $\frac{\text{change in } V}{\text{change in } t} = -5$ (e.g. $\frac{65-75}{5-3} = -5$, $\frac{40-65}{10-5} = -5$): the tank loses 5 L/min. (b) $V = -5t + 90$. (c) $V = -5(8) + 90 = 50$ L. (d) $-5t + 90 = 0$ gives $t = 18$ min. **15.** (a) A: $y = 3x + 1$; B: $y = -2x + 11$. (b) $3x + 1 = -2x + 11$ gives $x = 2$, and both give $y = 7$: the point $(2, 7)$. (c) Relation A — past $x = 2$ it gains 3 per step while B loses 2 per step. **16.** B. (The rate is $\frac{-7-11}{4-(-2)} = -3$, so $y = -3x + 5$; at $x = 1$, $y = 2$.) **17.** $k = 5$.

Lesson 2 — Extra Practice

Slope–Intercept Form: $y = mx + b$

Part A — Reading m and b When the Numbers Are Ugly

1. State the slope m and the y-intercept b of each equation. Each one is already solved for y, but the x-term is not always written first.

(a) $y = -\frac{1}{4}x + 6$

(b) $y = 7 - \frac{3}{2}x$

(c) $y = \frac{5x - 8}{2}$

(d) $y = -x$

(e) $y = 0.4x - 1.5$

(f) $y = \frac{2}{3} - \frac{x}{6}$

Part B — Rearrange First, Then Read

2. Rearrange each equation into slope–intercept form, then state m and b. Remember to divide *every* term by the coefficient of y.

(a) $2y = 6x - 8$

(b) $3x + y = 7$

(c) $y - 4 = 2x$

(d) $5x - 2y = 12$

(e) $3x + 4y = 0$

(f) $-x + 3y = 9$

3. Rearrange into slope–intercept form and state m and b. These take more than one step, and three of the four have a fractional slope or intercept.

(a) $6x - 4y = 10$

(b) $2(y - 3) = 5x$

(c) $\frac{y}{3} = x - 2$

(d) $10 - 4y = 6x$

Part C — Matching and Comparing Lines

4. Match each description on the right to exactly one equation on the left. Every equation is used once.

A. $y = 3x - 2$
B. $y = -3x + 2$
C. $y = \frac{1}{3}x + 2$
D. $y = -2$
E. $y = 3x + 2$

(i) Crosses the y-axis at $(0, 2)$ and falls 3 units for every step right. _____
(ii) A horizontal line passing 2 units below the origin. _____
(iii) Rises 3 units per step right and crosses the y-axis *below* the origin. _____
(iv) Rises only 1 unit for every 3 units right. _____
(v) Exactly as steep as A, but crosses the y-axis 4 units higher. _____

5. **Compare the pair.** For each pair, state (i) which line is **steeper** and how you know, and (ii) which line crosses the y-axis **higher** (or say they cross at the same point). Rearrange first where needed.

(a) $y = -5x + 1$ and $y = 4x - 3$ (b) $2y = 3x + 10$ and $y = -\frac{7}{4}x + 5$ (c) $y = \frac{2}{3}x - 4$ and $3x - 4y = 8$

Part D — Writing the Equation

6. Write the equation of the line in the form $y = mx + b$. Watch the two special cases.

(a) $m = -\frac{3}{4}$, $b = 5$ (b) $m = 2$, $b = 0$ (c) $m = 0$, $b = -6$

(d) $m = 1$, $b = -\frac{1}{2}$ (e) $m = -1$, $b = 0$ (f) Slope $\frac{5}{2}$, passing through $(0, -3)$

7. Explain in one or two sentences what is different about the *graph* in each special case above: (a) the line from part (b), where $b = 0$; and (b) the line from part (c), where $m = 0$.

Part E — Error Analysis

8. Devon looks at $3x + y = 7$ and writes "$m = 3$ and $b = 7$." Exactly one of those two values is correct. State which one, correct the other, and explain why this equation is so easy to misread.

9. Casey rearranges $2y = 6x - 8$ and writes $y = 6x - 4$. Identify the mistake in one sentence, then give the correct equation together with the correct m and b.

10. Sam rearranges $5x - 2y = 12$ like this: $-2y = -5x + 12$, then $y = \frac{5}{2}x + 6$. The slope is right but the y-intercept is wrong. Find the step where the error happens and write out the corrected final line.

Part F — Working Backwards

11. **Parallel lines have equal slopes.** Determine the required value in each case.

(a) Find m so that $y = mx + 4$ is parallel to $y = -2x + 9$.

(b) Find k so that $y = kx - 1$ is parallel to $3x + y = 5$.

(c) Find the slope of any line parallel to $4x - 2y = 7$.

(d) Find k so that $2y = kx + 6$ is parallel to $y = -5x + 1$.

12. Determine the missing value in each equation.

(a) $y = 3x + b$ passes through $(0, -7)$. Find b.

(b) $y = -2x + b$ passes through $(4, 1)$. Find b.

(c) $y = mx - 4$ passes through $(2, 6)$. Find m.

Part G — Reasoning, Application, and Stretch

13. Start with the line $y = 2x + 3$. Describe precisely what happens to its graph under each change — say whether the line tilts, slides, or both, and in which direction.

(a) b changes from 3 to -1. (b) m changes from 2 to 5. (c) m changes from 2 to -2. (d) m changes from 2 to $\frac{1}{2}$.

14. Two different lines have the same value of b but different values of m. What must the two graphs have in common, and what must be different? Sketch nothing — describe it in words.

15. Two different lines have the same value of m but different values of b. Explain why these two lines can never cross, no matter how far they are extended.

16. A taxi fare satisfies $4C = 3d + 26$, where d is the distance travelled in kilometres and C is the fare in dollars.

(a) Rewrite the equation in the form $C = md + b$.

(b) State the cost per kilometre and explain how you know.

(c) State the flat fee charged before any distance is driven.

(d) Determine the fare for a 12 km trip.

17. **Stretch.** Three equations are given below. Two of them describe the *same* line and one does not.
(i) $4x + 2y = 10$ (ii) $y - 5 = -2x$ (iii) $2x + y = -5$

Rearrange all three into slope–intercept form, identify the odd one out, and explain what the two matching equations have in common that the third one does not.

18. *(Multiple Choice)* Which equation has slope $-\dfrac{2}{3}$ and y-intercept 4?
A. $2x + 3y = 12$ B. $3x + 2y = 8$
C. $2x - 3y = 12$ D. $3x - 2y = -8$

19. *(Numerical Response)* For the line $6x - 3y = -15$, the value of $m + b$ is _____.

Answer Key. 1. (a) $m = -\frac{1}{4}$, $b = 6$ (b) $m = -\frac{3}{2}$, $b = 7$ (c) $y = \frac{5}{2}x - 4$: $m = \frac{5}{2}$, $b = -4$ (d) $m = -1$, $b = 0$ (e) $m = 0.4$, $b = -1.5$ (f) $m = -\frac{1}{6}$, $b = \frac{2}{3}$. **2.** (a) $y = 3x - 4$; $m = 3$, $b = -4$ (b) $y = -3x + 7$; $m = -3$, $b = 7$ (c) $y = 2x + 4$; $m = 2$, $b = 4$ (d) $y = \frac{5}{2}x - 6$; $m = \frac{5}{2}$, $b = -6$ (e) $y = -\frac{3}{4}x$; $m = -\frac{3}{4}$, $b = 0$ (f) $y = \frac{1}{3}x + 3$; $m = \frac{1}{3}$, $b = 3$. **3.** (a) $y = \frac{3}{2}x - \frac{5}{2}$; $m = \frac{3}{2}$, $b = -\frac{5}{2}$ (b) $y = \frac{5}{2}x + 3$; $m = \frac{5}{2}$, $b = 3$ (c) $y = 3x - 6$; $m = 3$, $b = -6$ (d) $y = -\frac{3}{2}x + \frac{5}{2}$; $m = -\frac{3}{2}$, $b = \frac{5}{2}$. **4.** (i) B (ii) D (iii) A (iv) C (v) E. **5.** (a) Steeper: $y = -5x + 1$, since $|-5| = 5 > 4$; it also crosses higher ($1 > -3$). (b) $2y = 3x + 10$ gives $y = \frac{3}{2}x + 5$; steeper is $y = -\frac{7}{4}x + 5$ since $1.75 > 1.5$; both cross at the same point $(0, 5)$. (c) $3x - 4y = 8$ gives $y = \frac{3}{4}x - 2$; it is steeper ($\frac{3}{4} > \frac{2}{3}$) and it also crosses higher ($-2 > -4$). **6.** (a) $y = -\frac{3}{4}x + 5$ (b) $y = 2x$ (c) $y = -6$ (d) $y = x - \frac{1}{2}$ (e) $y = -x$ (f) $y = \frac{5}{2}x - 3$. **7.** (a) With $b = 0$ the line passes through the origin $(0, 0)$. (b) With $m = 0$ there is no rise at all, so the graph is a horizontal line through $(0, -6)$. **8.** $b = 7$ is correct; m is -3, not 3. Rearranged, $y = -3x + 7$ — moving $3x$ to the right side changes its sign, and the equation looks like it already shows m and b even though y was never isolated. **9.** She divided only the constant term by 2; every term must be divided. Correctly, $y = 3x - 4$, so $m = 3$ and $b = -4$. **10.** The error is in dividing by -2: the $+12$ must also change sign, giving -6, not $+6$. Correct: $y = \frac{5}{2}x - 6$. **11.** (a) $m = -2$ (b) $3x + y = 5$ gives $y = -3x + 5$, so $k = -3$ (c) $4x - 2y = 7$ gives $y = 2x - \frac{7}{2}$, so the slope is 2 (d) $2y = kx + 6$ gives $y = \frac{k}{2}x + 3$, so $\frac{k}{2} = -5$ and $k = -10$. **12.** (a) $b = -7$ (b) $1 = -2(4) + b$, so $b = 9$ (c) $6 = 2m - 4$, so $m = 5$. **13.** (a) Same tilt, slides down 4 units; it now crosses at $(0, -1)$. (b) Same crossing point $(0, 3)$, but the line tilts steeper upward. (c) Same crossing point, but the line now falls to the right instead of rising, at the same steepness. (d) Same crossing point, but the tilt flattens — it still rises, just more gently. **14.** They cross the y-axis at the same point $(0, b)$, so they intersect there; their steepness (and possibly their direction) differs, so they spread apart away from that point. **15.** Equal slopes means both lines climb at exactly the same rate, so the vertical gap between them stays $b_1 - b_2$ forever; since that gap is not zero, they are parallel and never meet. **16.** (a) $C = \frac{3}{4}d + \frac{13}{2}$, i.e. $C = 0.75d + 6.5$. (b) \$0.75 per km — it is the slope, the extra cost for each additional kilometre. (c) \$6.50, the fare when $d = 0$. (d) $C = 0.75(12) + 6.5 = \$\mathbf{15.50}$. **17.** (i) $y = -2x + 5$; (ii) $y = -2x + 5$; (iii) $y = -2x - 5$. The odd one out is (iii). Equations (i) and (ii) share both $m = -2$ and $b = 5$, so they are the identical line; (iii) has the same slope but a different intercept, so it is a parallel line, not the same one. **18.** A ($2x + 3y = 12$ gives $y = -\frac{2}{3}x + 4$). **19.** $6x - 3y = -15$ gives $y = 2x + 5$, so $m + b = 2 + 5 = 7$.

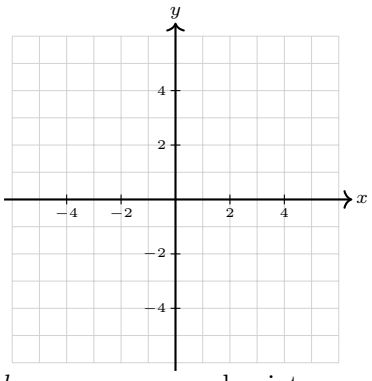
Lesson 3 — Extra Practice

Graphing a Line Using Slope-Intercept Form

Part A — Counting Rise Over Run: Fractional Slopes

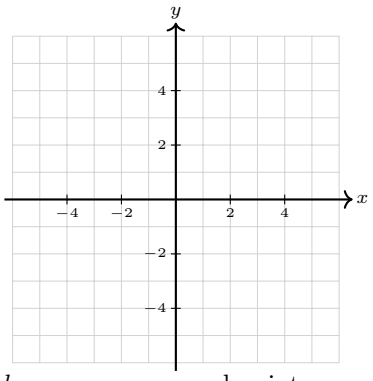
1. Graph each line. Plot $(0, b)$ first, then use the slope to step to a second point, then draw the line all the way across the grid. Record the second point you landed on.

(a) $y = \frac{2}{3}x - 2$



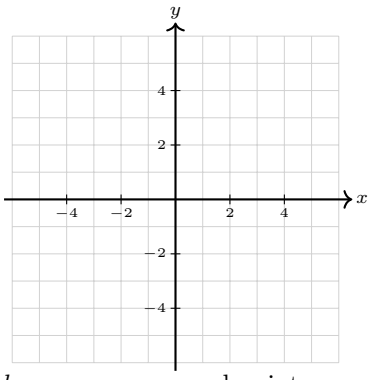
$b =$_____ second point _____

(b) $y = -\frac{3}{4}x + 3$



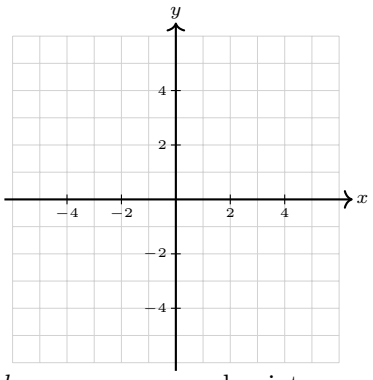
$b =$_____ second point _____

(c) $y = \frac{5}{2}x - 4$



$b =$_____ second point _____

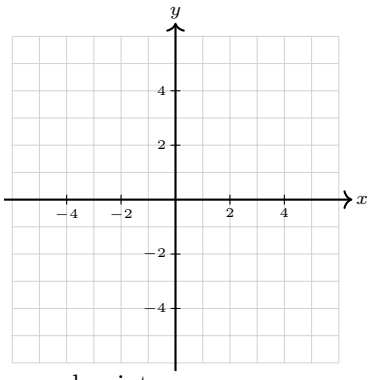
(d) $y = -\frac{4}{3}x + 1$



$b =$_____ second point _____

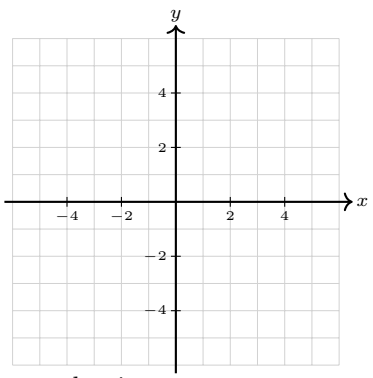
2. **One line, two directions.** Both grids below are for the same line, $y = -\frac{2}{3}x + 1$.

(a) Write the slope as $\frac{-2}{3}$. From $(0, 1)$, step **right** 3 and **down** 2. Plot that point and draw the line.



second point _____

(b) Write the same slope as $\frac{2}{-3}$. From $(0, 1)$, step **left** 3 and **up** 2. Plot that point and draw the line.



second point _____

(c) The two second points are different, but the two lines are identical. Explain why.

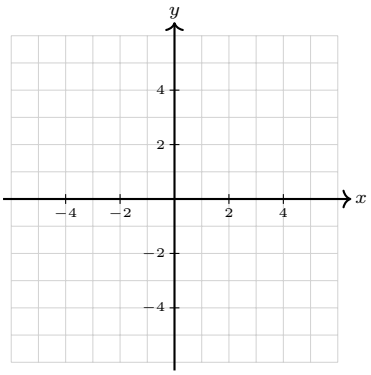
3. Write each slope as *two* equivalent fractions $\frac{\text{rise}}{\text{run}}$: one that steps to the **right**, and one that steps to the **left**.

(a) $m = -4$ (b) $m = -\frac{3}{5}$ (c) $m = \frac{7}{2}$

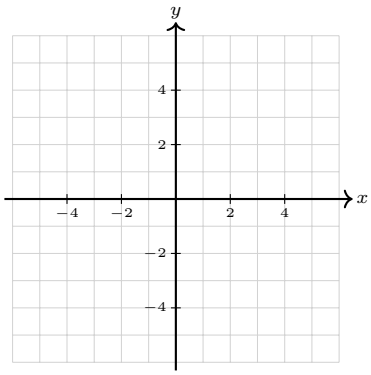
Part B — Rearrange First, Then Graph

4. None of these equations is in $y = mx + b$ form yet. Rearrange each one, write the rearranged equation on the line provided, then graph it.

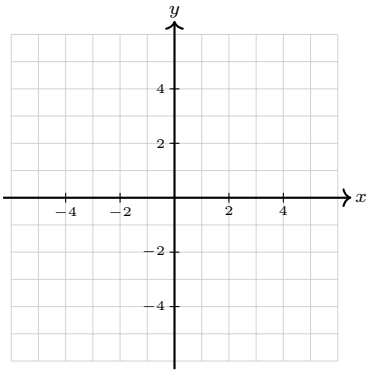
(a) $2y = 6x - 4$ $\Rightarrow$ _____



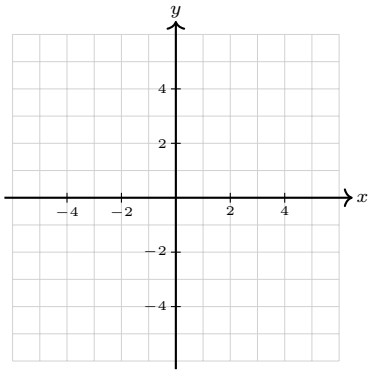
(b) $3x + y = 5$ $\Rightarrow$ _____



(c) $3x + 4y = 8$ $\Rightarrow$ _____



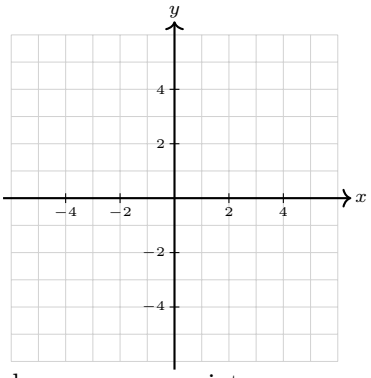
(d) $x - 2y = 6$ $\Rightarrow$ _____



Part C — Horizontal and Vertical Lines

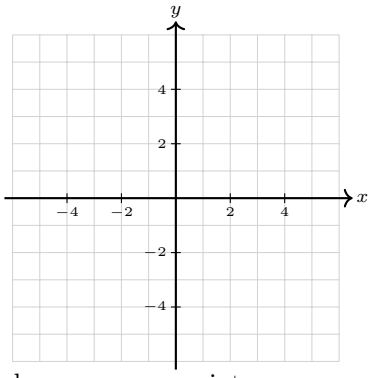
5. Graph each line, then fill in all three blanks. Write *none* or *undefined* where that is the correct answer.

(a) $y = -3$



slope _____ y-int _____ x-int _____

(b) $x = 4$



slope _____ y-int _____ x-int _____

6. One of the two lines in Question 5 can be written in the form $y = mx + b$ and the other cannot. State which is which, write the one that can in $y = mx + b$ form, and explain what goes wrong when you try to force the other one into that form.

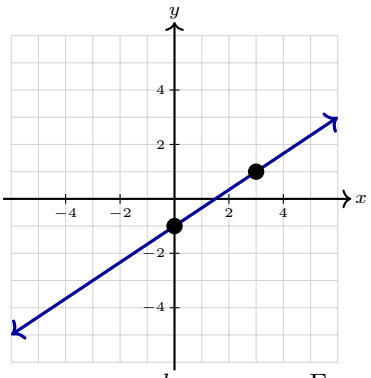
7. For each equation, state the slope and then answer **yes** or **no**: can it be written in the form $y = mx + b$?

(a) $x = 0$ (the y-axis) (b) $y = 0$ (the x-axis) (c) $x = 3.5$

Part D — Reading the Equation Off a Graph

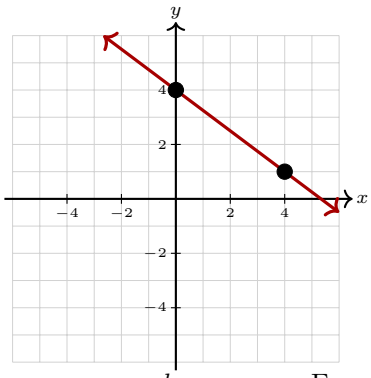
8. Each line below passes through the two marked lattice points. Read the slope and the y-intercept from the graph, then write the equation in $y = mx + b$ form.

(a)



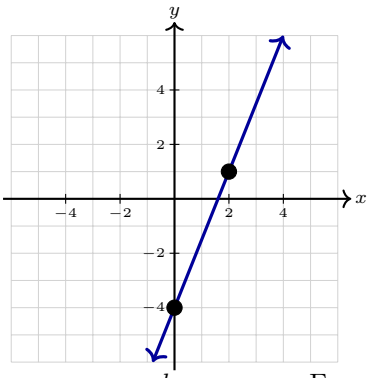
$m =$_____ $b =$_____ Equation: _____

(b)



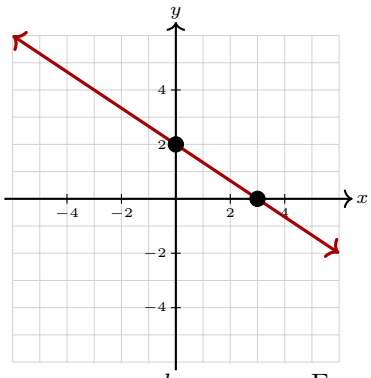
$m =$_____ $b =$_____ Equation: _____

(c)



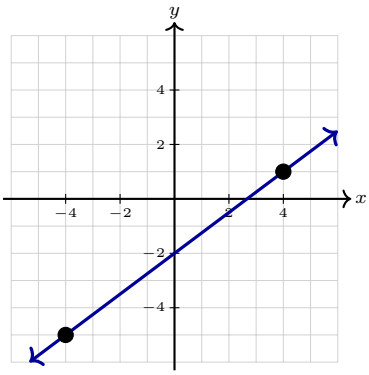
$m =$_____ $b =$_____ Equation: _____

(d)



$m =$_____ $b =$_____ Equation: _____

9. The graph below shows a single line through the two marked points. Use it for parts (a) to (e).



(a) Determine the slope by counting rise over run between the two marked points.
(b) State the y-intercept as an ordered pair.
(c) Write the equation of the line.
(d) Give the coordinates of a third point on the line, different from the two marked ones.
(e) Decide whether each point lies on the line, and show the check: $(6, 3)$ and $(8, 4)$.

Part E — Error Analysis

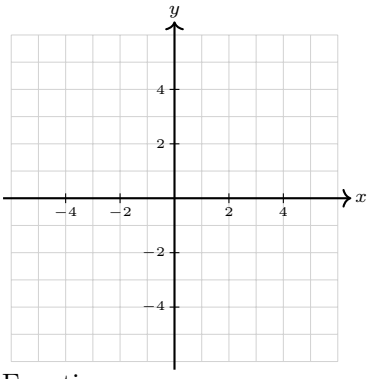
10. Marcus graphs $y = \frac{2}{3}x + 1$. He plots $(0, 1)$ correctly, then moves *right* 2 *and up* 3 and plots $(2, 4)$. Explain the mistake, state the slope of the line he actually drew, and give the correct second point.

11. Priya graphs $y = -\frac{2}{5}x + 3$. She plots $(0, 3)$, then moves *right* 5 *and up* 2 and plots $(5, 5)$. Explain her mistake and give the correct point she should have reached by moving right 5. Then give a second correct point found by moving *left* instead.

Part F — Working Backwards: The Point Is Not the Intercept

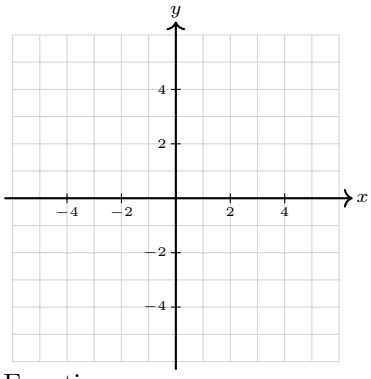
12. In each part you are given **one point on the line** (which is *not* the y-intercept) and the slope. Plot the given point, use the slope to step to more points, draw the line, and then write its equation in $y = mx + b$ form.

(a) Through $(-2, 3)$ with $m = \dfrac{1}{2}$



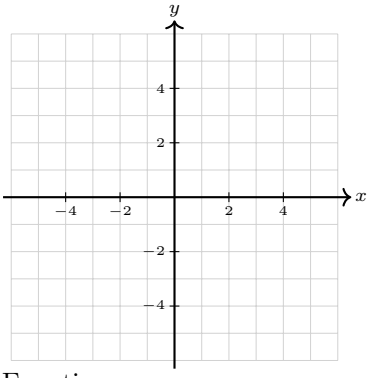
Equation: _____

(b) Through $(3, -2)$ with $m = -\dfrac{2}{3}$



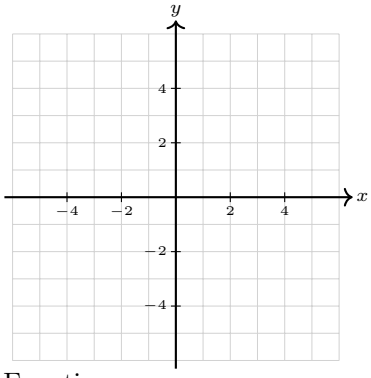
Equation: _____

(c) Through $(-4, -1)$ with $m = \dfrac{5}{4}$



Equation: _____

(d) Through $(2, -3)$ with $m = 0$



Equation: _____

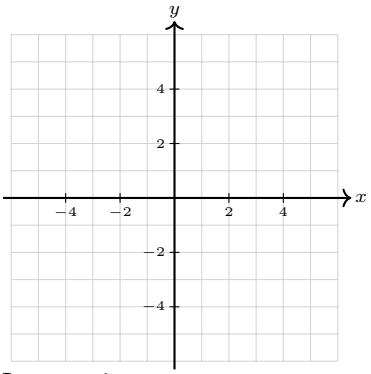
Part G — Reasoning, Stretch, and Multiple Choice

13. Without graphing, decide whether the point $(6, 1)$ lies **above**, **below**, or **on** the line $y = -\dfrac{2}{3}x + 4$. Justify your answer with a calculation.

14. A line has slope $\dfrac{2}{5}$ and y-intercept -1. The grid only reaches $x = 6$. Explain, using repeated rise-over-run steps, how you know $(10, 3)$ is on this line without extending the grid.

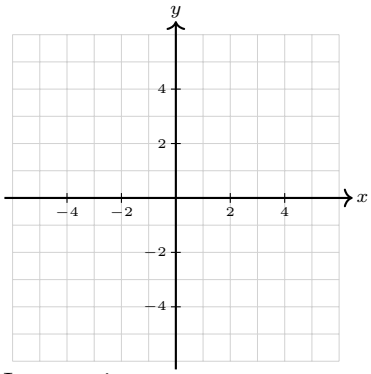
15. **Stretch.** Graph both lines on the same grid. Read the point of intersection from your graph, then confirm it algebraically by setting the two expressions for y equal and solving.

(a) $y = 2x - 3$ and $y = -\frac{1}{2}x + 2$



Intersection: _____

(b) $y = \frac{1}{3}x + 1$ and $y = -x + 5$



Intersection: _____

Algebraic confirmation for (a) and (b):

16. **Stretch.** Devon graphs $y = \frac{4}{6}x - 1$ by stepping right 6 and up 4. Elena graphs the same equation by first reducing the slope to $\frac{2}{3}$ and stepping right 3 and up 2. They land on different second points. Explain why both students end up drawing exactly the same line, and state one advantage of Elena's method.

17. *(Multiple Choice)* Starting at the y-intercept of $y = -\frac{5}{3}x + 2$, which move lands on another point of the line?

A. Right 3, up 5 B. Right 5, down 3
C. Left 5, up 3 D. Right 3, down 5

18. *(Numerical Response)* The line $y = \frac{2}{3}x - 4$ passes through the point $(k, 2)$. The value of k is _____.

Answer Key. 1. (a) $b = -2$, right 3 up 2 to $(3, 0)$ (b) $b = 3$, right 4 down 3 to $(4, 0)$ (c) $b = -4$, right 2 up 5 to $(2, 1)$ (d) $b = 1$, right 3 down 4 to $(3, -3)$. **2.** (a) $(3, -1)$ (b) $(-3, 3)$ (c) $\frac{-2}{3} = \frac{2}{-3}$, so both moves have the same slope; each point is on the line $y = -\frac{2}{3}x + 1$ (check: $-\frac{2}{3}(3) + 1 = -1$ and $-\frac{2}{3}(-3) + 1 = 3$). Two points determine one line, and both pairs of points sit on that same line — going down-and-right or up-and-left just travels the line in opposite directions. **3.** (a) $\frac{-4}{1}$ (right 1, down 4) and $\frac{4}{-1}$ (left 1, up 4) (b) $\frac{-3}{5}$ and $\frac{3}{-5}$ (c) $\frac{7}{2}$ (right 2, up 7) and $\frac{-7}{-2}$ (left 2, down 7). **4.** (a) $y = 3x - 2$; from $(0, -2)$ right 1 up 3 to $(1, 1)$ (b) $y = -3x + 5$; from $(0, 5)$ right 1 down 3 to $(1, 2)$ (c) $y = -\frac{3}{4}x + 2$; from $(0, 2)$ right 4 down 3 to $(4, -1)$ (d) $y = \frac{1}{2}x - 3$; from $(0, -3)$ right 2 up 1 to $(2, -2)$. **5.** (a) $y = -3$: horizontal through $(0, -3)$; slope 0, y-int -3, x-int none (it never meets the x-axis). (b) $x = 4$: vertical through $(4, 0)$; slope undefined, y-int none, x-int 4. **6.** $y = -3$ can: it is $y = 0x - 3$, with $m = 0$ and $b = -3$. $x = 4$ cannot. Every equation $y = mx + b$ gives exactly one y for each x, but $x = 4$ pairs the single input 4 with every possible y — and there is no slope m to write, since the run between any two of its points is 0 and division by 0 is undefined. **7.** (a) slope undefined; **no** (vertical). (b) slope 0; **yes**, $y = 0x + 0$. (c) slope undefined; **no** (vertical). **8.** (a) $m = \frac{2}{3}$, $b = -1$, $y = \frac{2}{3}x - 1$ (b) $m = -\frac{3}{4}$, $b = 4$, $y = -\frac{3}{4}x + 4$ (c) $m = \frac{5}{2}$, $b = -4$, $y = \frac{5}{2}x - 4$ (d) $m = -\frac{2}{3}$, $b = 2$, $y = -\frac{2}{3}x + 2$. **9.** (a) from $(-4, -5)$ to $(4, 1)$: rise 6, run 8, $m = \frac{6}{8} = \frac{3}{4}$ (b) $(0, -2)$ (c) $y = \frac{3}{4}x - 2$ (d) $(0, -2)$ works, and so does any point such as $(8, 4)$ or $(-8, -8)$ (e) $(6, 3)$: $\frac{3}{4}(6) - 2 = \frac{5}{2} = 2.5 \neq 3$, so **not** on the line; $(8, 4)$: $\frac{3}{4}(8) - 2 = 4$, so it **is** on the line. **10.** He counted the run before the rise, using the numerator as the run. In $\frac{\text{rise}}{\text{run}}$ the top number is the rise, so $\frac{2}{3}$ means right 3, up 2. The line he drew has slope $\frac{3}{2}$ (too steep). Correct second point: $(3, 3)$. **11.** A negative slope falls to the right, so after moving right she must move *down*, not up: $\frac{-2}{5}$ means right 5, down 2, giving $(5, 1)$. Moving left instead reverses both signs, $\frac{2}{-5}$: left 5, up 2, giving $(-5, 5)$. Check: $-\frac{2}{5}(5) + 3 = 1$

and $-\frac{2}{5}(-5)+3=5$; her point $(5,5)$ is not on the line. **12.** (a) $y=\frac{1}{2}x+4$ (from $(-2,3)$ step right 2 up 1 to $(0,4)$, the intercept) (b) $y=-\frac{2}{3}x$ (from $(3,-2)$ step left 3 up 2 to $(0,0)$) (c) $y=\frac{5}{4}x+4$ (from $(-4,-1)$ step right 4 up 5 to $(0,4)$) (d) $y=-3$, a horizontal line through $(2,-3)$. **13.** At $x=6$, $y=-\frac{2}{3}(6)+4=0$, so the line passes through $(6,0)$. The point $(6,1)$ has a greater y-value at the same x, so it lies **above** the line. **14.** Start at $(0,-1)$. Each step of right 5, up 2 stays on the line: $(5,1)$, then $(10,3)$. Two steps land exactly on $(10,3)$, so it is on the line. Check: $\frac{2}{5}(10)-1=4-1=3$. **15.** (a) $(2,1)$: $2x-3=-\frac{1}{2}x+2 \Rightarrow \frac{5}{2}x=5 \Rightarrow x=2$, and $y=2(2)-3=1$. (b) $(3,2)$: $\frac{1}{3}x+1=-x+5 \Rightarrow \frac{4}{3}x=4 \Rightarrow x=3$, and $y=-3+5=2$. **16.** $\frac{4}{6}=\frac{2}{3}$, so the two slopes are the same number and both students start at the same intercept $(0,-1)$. Devon lands on $(6,3)$ and Elena on $(3,1)$, but both of those points satisfy $y=\frac{2}{3}x-1$, and a line through $(0,-1)$ with that slope is unique — so the lines coincide. Elena's advantage: the smaller numbers keep her second point on the grid (and she can repeat the step several times to get extra points for accuracy). **17.** D (right 3, down 5 gives $(3,-3)$, and $-\frac{5}{3}(3)+2=-3$). **18.** $k=9$ (since $\frac{2}{3}(9)-4=6-4=2$).

Lesson 4 — Extra Practice

Writing the Equation of a Line Given Slope and y-Intercept

Part A — Slopes and Intercepts That Are Not Whole Numbers

1. Write the equation of each line in the form $y = mx + b$. Keep every fraction exact.

(a) slope $-\frac{3}{4}$, y-intercept $\frac{5}{2}$

(b) slope $\frac{7}{5}$, y-intercept $-\frac{1}{2}$

(c) slope -6, y-intercept $-\frac{2}{3}$

(d) slope $\frac{1}{8}$, y-intercept -9

(e) slope $-\frac{5}{6}$, through $\left(0, \frac{11}{3}\right)$

(f) y-intercept $-\frac{7}{4}$, slope $-\frac{9}{2}$

2. Write the equation of each line. Each one has a slope of 0, a y-intercept of 0, or both.

(a) slope 0, y-int -5

(b) slope $-\frac{2}{7}$, y-int 0

(c) slope 0, through $\left(0, \frac{9}{2}\right)$

(d) slope $\frac{4}{3}$, through the origin

3. Answer each question in full sentences where asked.

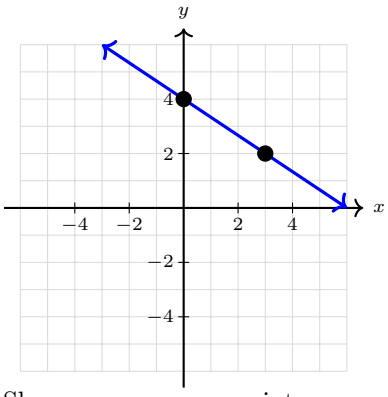
(a) A line has slope 0 and y-intercept -5. Explain why its equation contains no x term, then list the coordinates of three different points that lie on it.

(b) A line has y-intercept 0 and slope $-\frac{2}{7}$. Explain why $(0, 0)$ must be on the line, then determine the y-value when $x = 14$.

Part B — Reading a Line from a Graph or a Description

4. Each line passes through the two marked lattice points. Read the slope and the y-intercept from the graph, then write the equation.

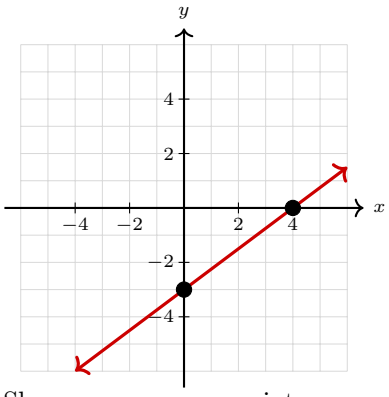
(a)



Slope: _____ y-int: _____

Equation: _____

(b)

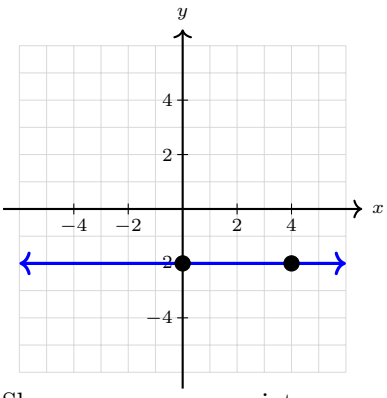


Slope: _____ y-int: _____

Equation: _____

5. Write the equation of each line. One of them is a special case — think about what its slope must be.

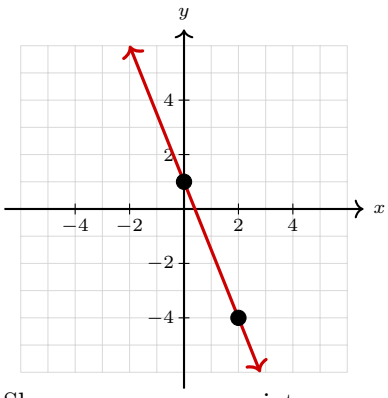
(a)



Slope: _____ y-int: _____

Equation: _____

(b)



Slope: _____ y-int: _____

Equation: _____

6. Each description gives a y-intercept and a rate of change in words. Write the equation of the line.

(a) The line crosses the y-axis at -3 and falls 2 units for every 5 units across.

(b) The line passes through the origin and rises 7 units for every 4 units across.

(c) The line is horizontal and crosses the y-axis at $\frac{5}{2}$.

(d) The line crosses the y-axis at 6 and falls 9 units for every 3 units across.

7. **Work backwards.** Each equation is given. Describe the line in words: state the point where it crosses the y-axis, and say how far it rises or falls for a stated run.

(a) $y = -\frac{3}{5}x + 7$

(b) $y = \frac{9}{2}x$

8. **Error analysis.** Find the mistake, explain it, and give the correct equation.

(a) Asked for the line with slope -4 and y-intercept $\frac{3}{2}$, Tomas wrote $y = \frac{3}{2}x - 4$. Explain his error, write the correct equation, then evaluate both equations at $x = 2$ to show they are different lines.

(b) Told that a line has slope 0 and y-intercept -7, Priya wrote $x = -7$. Explain what is wrong with her answer, describe what the graph of $x = -7$ actually looks like, and give the correct equation.

Part C — Applications

9. A bicycle courier charges a flat dispatch fee of \$8.50 plus \$1.25 per kilometre. Let C be the total charge in dollars for a delivery of d kilometres.

(a) Write an equation for C in terms of d.

(b) Determine the charge for a 14 km delivery.

(c) What does the ordered pair $(0, 8.50)$ mean in this situation?

10. A tutoring service charges a one-time registration fee of \$36, then bills \$150 for every 4 sessions. Let C be the total cost in dollars after n sessions.

(a) Determine the cost of a single session.

(b) Write an equation for C in terms of n.

(c) Determine the total cost of 10 sessions.

11. A swimming pool holds 4200 L of water when the drain is opened. The water leaves at a constant rate of 175 L per hour. Let V be the volume of water in litres after t hours.

(a) Write an equation for V in terms of t.

(b) Determine the volume remaining after 6 hours.

(c) Determine how long the pool takes to empty completely.

12. A stamping machine starts empty and produces parts at a constant rate of 45 parts every 6 minutes. Let N be the number of parts produced after t minutes.

(a) Determine the number of parts produced per minute, as a fraction in lowest terms.

(b) Write an equation for N in terms of t, and explain why $b = 0$ here.

(c) Determine the number of parts produced in 40 minutes.

Part D — Reasoning and Stretch

13. **Stretch.** Consider the three lines $y = 3x - 4$, $y = -\frac{1}{2}x - 4$, and $y = -4$.

(a) What do all three graphs have in common? Name the exact point they share.

(b) Describe how the three graphs differ from one another.

(c) A fourth line shares that same point and has slope $\frac{5}{2}$. Write its equation.

14. The lines $y = \frac{2}{3}x + 1$ and $y = \frac{2}{3}x - 5$ have the same slope but different y-intercepts.

(a) What is the relationship between the two graphs? How many points do they share?

(b) Determine the y-value on each line when $x = 6$, and state the vertical distance between the two lines at that x-value.

15. **Stretch.** A line crosses the y-axis at -6. It is exactly as steep as the line $y = -\frac{7}{4}x + 2$, but it rises from left to right instead of falling. Write its equation, then determine its y-value at $x = 8$.

16. *(MC)* Which equation has slope $-\frac{5}{3}$ and y-intercept $\frac{4}{3}$?

A. $y = \frac{4}{3}x - \frac{5}{3}$ B. $y = -\frac{5}{3}x + \frac{4}{3}$

C. $y = \frac{5}{3}x - \frac{4}{3}$ D. $y = -\frac{4}{3}x + \frac{5}{3}$

17. *(NR)* A line has slope $-\frac{3}{4}$ and y-intercept 9. Its y-value when $x = 8$ is _____.

Answer Key. 1. (a) $y = -\frac{3}{4}x + \frac{5}{2}$ (b) $y = \frac{7}{5}x - \frac{1}{2}$ (c) $y = -6x - \frac{2}{3}$ (d) $y = \frac{1}{8}x - 9$ (e) $y = -\frac{5}{6}x + \frac{11}{3}$ (f) $y = -\frac{9}{2}x - \frac{7}{4}$. **2.** (a) $y = -5$ (b) $y = -\frac{2}{7}x$ (c) $y = \frac{9}{2}$ (d) $y = \frac{4}{3}x$. **3.** (a) $m = 0$, so the term mx is $0 \cdot x = 0$ and disappears; every point has $y = -5$, e.g. $(-2, -5)$, $(0, -5)$, $(3, -5)$. (b) $b = 0$ means the line meets the y-axis at height 0, which is the origin; at $x = 14$, $y = -\frac{2}{7}(14) = -4$. **4.** (a) $m = -\frac{2}{3}$, $b = 4$, $y = -\frac{2}{3}x + 4$ (b) $m = \frac{3}{4}$, $b = -3$, $y = \frac{3}{4}x - 3$. **5.** (a) $m = 0$, $b = -2$, $y = -2$ (b) $m = -\frac{5}{2}$, $b = 1$, $y = -\frac{5}{2}x + 1$. **6.** (a) $y = -\frac{2}{5}x - 3$ (b) $y = \frac{7}{4}x$ (c) $y = \frac{5}{2}$ (d) $y = -3x + 6$. **7.** (a) It crosses the y-axis at $(0, 7)$ and falls 3 units for every 5 units right. (b) It passes through the origin $(0, 0)$ and rises 9 units for every 2 units right. **8.** (a) He swapped m and b; correctly $y = -4x + \frac{3}{2}$. At $x = 2$ his equation gives $3 - 4 = -1$, the correct one gives $-8 + \frac{3}{2} = -\frac{13}{2}$, so they are different lines. (b) Slope 0 means horizontal, and $x = -7$ is a *vertical* line (every point has $x = -7$) whose slope is undefined; the correct equation is $y = -7$. **9.** (a) $C = 1.25d + 8.50$ (b) \$26.00 (c) with a 0 km delivery the charge is still \$8.50 — the flat dispatch fee. **10.** (a) $\frac{150}{4} = \$37.50$ per session (b) $C = 37.50n + 36$ (c) \$411. **11.** (a) $V = -175t + 4200$ (b) 3150 L (c) $4200 \div 175 = 24$ hours. **12.** (a) $\frac{45}{6} = \frac{15}{2}$ parts per minute (b) $N = \frac{15}{2}t$; the machine starts empty, so 0 parts exist at $t = 0$ (c) $\frac{15}{2}(40) = 300$ parts. **13.** (a) All three have $b = -4$, so all three pass through $(0, -4)$. (b) They differ in slope: $y = 3x - 4$ rises steeply, $y = -\frac{1}{2}x - 4$ falls gently, $y = -4$ is horizontal. (c) $y = \frac{5}{2}x - 4$. **14.** (a) Equal slopes means the lines are parallel; they share no points. (b) At $x = 6$: $y = 5$ and $y = -1$; the vertical gap is 6 units (the same as the gap between the intercepts). **15.** Equally steep but rising gives $m = +\frac{7}{4}$, so $y = \frac{7}{4}x - 6$; at $x = 8$, $y = 14 - 6 = 8$. **16.** B. **17.** 3.

Lesson 5 — Extra Practice

Writing the Equation of a Line Given Slope and a Point

Part A — Substitute and Solve with Harder Numbers

1. Determine the equation of each line in the form $y = mx + b$. None of the given points is the y-intercept, so each one needs the substitute-and-solve step.

(a) slope 6, point $(-3, -20)$

(b) slope -7, point $(-2, 9)$

(c) slope 4, point $(-5, 12)$

(d) slope -9, point $(3, -11)$

(e) slope 8, point $(-4, -3)$

(f) slope -6, point $(-7, 5)$

2. Determine the equation of each line. The slopes are fractions — keep them exact.

(a) slope $\frac{2}{3}$, point $(9, -1)$

(b) slope $-\frac{3}{5}$, point $(-10, 4)$

(c) slope $-\frac{7}{4}$, point $(-8, 3)$

(d) slope $\frac{5}{6}$, point $(-12, -7)$

3. Determine the equation of each line. Here mx does *not* divide evenly, so b itself is a fraction. Leave your answer exact.

(a) slope $\frac{1}{2}$, point $(5, -3)$

(b) slope $-\frac{2}{3}$, point $(4, 1)$

Part B — Horizontal, Vertical, Parallel and Perpendicular

4. Write the equation of each line. Two of them are not in the form $y = mx + b$ at all.

(a) slope 0, through $(-6, 5)$

(b) undefined slope, through $(-6, 5)$

(c) slope 0, through $(8, -9)$

(d) undefined slope, through $(-3, -4)$; explain in one sentence why this line has no $y = mx + b$ form.

5. **Parallel lines.** Parallel lines have equal slopes. Read the slope off the given line, then use the given point to find b.

(a) parallel to $y = 3x - 8$, through $(-2, 5)$

(b) parallel to $y = -\frac{3}{4}x + 1$, through $(8, -3)$

(c) parallel to $y = -\frac{5}{2}x$, through $(-4, 7)$

(d) parallel to $y = -6$, through $\left(3, \frac{11}{2}\right)$

6. **Perpendicular lines.** Perpendicular lines have slopes that are negative reciprocals. Find the new slope first, then use the given point to find b.

(a) perpendicular to $y = 2x + 9$, through $(4, -1)$

(b) perpendicular to $y = -\frac{3}{5}x + 2$, through $(-6, 4)$

(c) perpendicular to $y = -4x - 7$, through $(-8, 3)$

(d) perpendicular to $y = \frac{7}{2}x$, through $(14, -5)$

7. A horizontal and a vertical line are perpendicular to each other, but the negative-reciprocal rule cannot be used on them. Write the equation of each line described below.

(a) perpendicular to $y = 4$, through $(-5, 2)$

(b) perpendicular to $x = 3$, through $(-5, 2)$

Part C — Applications

8. A tank is draining at a constant rate of 5 L per minute. After 3 minutes of draining it holds 40 L. Let V be the volume in litres after t minutes.

(a) Write an equation for V in terms of t.

(b) How much water was in the tank when draining began?

(c) Determine when the tank is empty.

9. A hot tub cools at a constant rate of 4°F per hour once the heater is shut off. After 5 hours the water reads 82°F. Let T be the temperature after h hours.

(a) Write an equation for T in terms of h.

(b) What was the temperature when the heater was shut off?

(c) Determine when the water reaches 70°F.

10. A candle burns down at a constant rate of $\frac{5}{2}$ cm per hour. After 6 hours it is 9 cm tall. Let h be the height in centimetres after t hours.

(a) Write an equation for h in terms of t.

(b) Determine the original height of the candle.

(c) Determine when the candle burns out. Give an exact answer.

11. A gym charges a one-time joining fee plus a fixed monthly rate of \$40. A member who has belonged for 7 months has paid \$345 in total. Let C be the total paid after t months.

(a) Write an equation for C in terms of t.

(b) Determine the joining fee.

(c) Determine the total paid after 2 years.

Part D — Error Analysis, Working Backwards and Stretch

12. **Error analysis.** Find the mistake, explain it, and give the correct equation.

(a) Devon was asked for the line with slope -3 through $(5, 2)$. He wrote $5 = -3(2) + b$, found $b = 11$, and answered $y = -3x + 11$. Explain his error, then show that his equation fails the check at $(5, 2)$ and give the correct equation.

(b) Nina was asked for the line with slope $\frac{1}{2}$ through $(7, 3)$. She used the point's y-coordinate as b and wrote $y = \frac{1}{2}x + 3$. Explain why the y-coordinate of a point is not automatically b, then give the correct equation.

13. **Work backwards.** For each line, write the equation first, then determine where the line crosses *both* axes. Give exact coordinates.

(a) slope $-\frac{3}{4}$, through $(8, -1)$

(b) slope 5, through $(-3, -7)$

14. **Work backwards.** Each line is described by an unknown. Set up an equation and solve for it.

(a) A line passes through $(2, -5)$ and has y-intercept -11. Determine its slope m.

(b) The line $y = -\frac{2}{3}x + 10$ passes through the point $(a, 4)$. Determine the value of a.

15. **Stretch.** A line has slope $-\frac{2}{5}$ and passes through $(10, -1)$.

(a) Write its equation.

(b) Write the equation of the line parallel to it through $(-5, 8)$.

(c) Write the equation of the line perpendicular to it through $(10, -1)$.

16. *(MC)* Which is the equation of the line with slope -5 through $(-2, 7)$?

A. $y = -5x - 3$ B. $y = -5x + 17$
C. $y = -5x + 3$ D. $y = 5x - 3$

17. *(NR)* A line perpendicular to $y = -\frac{5}{2}x + 1$ passes through $(10, 9)$. Written in the form $y = mx + b$, the value of b is _____.

Answer Key. 1. (a) $y = 6x - 2$ (b) $y = -7x - 5$ (c) $y = 4x + 32$ (d) $y = -9x + 16$ (e) $y = 8x + 29$ (f) $y = -6x - 37$. **2.** (a) $y = \frac{2}{3}x - 7$ (b) $y = -\frac{3}{5}x - 2$ (c) $y = -\frac{7}{4}x - 11$ (d) $y = \frac{5}{6}x + 3$. **3.** (a) $-3 = \frac{5}{2} + b$, $b = -\frac{11}{2}$: $y = \frac{1}{2}x - \frac{11}{2}$ (b) $1 = -\frac{8}{3} + b$, $b = \frac{11}{3}$: $y = -\frac{2}{3}x + \frac{11}{3}$. **4.** (a) $y = 5$ (b) $x = -6$ (c) $y = -9$ (d) $x = -3$; a vertical line's slope is undefined, so there is no value of m to put into $y = mx + b$ (and every point shares one x-value, not one y-value). **5.** (a) $y = 3x + 11$ (b) $y = -\frac{3}{4}x + 3$ (c) $y = -\frac{5}{2}x - 3$ (d) $y = \frac{11}{2}$. **6.** (a) $m = -\frac{1}{2}$: $y = -\frac{1}{2}x + 1$ (b) $m = \frac{5}{3}$: $y = \frac{5}{3}x + 14$ (c) $m = \frac{1}{4}$: $y = \frac{1}{4}x + 5$ (d) $m = -\frac{2}{7}$: $y = -\frac{2}{7}x - 1$. **7.** (a) $x = -5$ (b) $y = 2$. **8.** (a) $40 = -5(3) + b$, $b = 55$: $V = -5t + 55$ (b) 55 L (c) $0 = -5t + 55$, $t = 11$ min. **9.** (a) $T = -4h + 102$ (b) 102°F (c) solve $70 = -4h + 102$: $h = 8$ hours. **10.** (a) $h = -\frac{5}{2}t + 24$ (b) 24 cm (c) $t = \frac{48}{5} = 9.6$ hours. **11.** (a) $345 = 40(7) + b$, $b = 65$: $C = 40t + 65$ (b) \$65 (c) 24 months: \$1025. **12.** (a) He substituted the coordinates backwards (x into y's place). Checking his answer: $-3(5) + 11 = -4 \neq 2$. Correctly $2 = -3(5) + b$, $b = 17$, so $y = -3x + 17$. (b) b is the y-value only at $x = 0$; the point $(7, 3)$ is not on the y-axis, so 3 is not b. Correctly $3 = \frac{1}{2}(7) + b$, $b = -\frac{1}{2}$, so $y = \frac{1}{2}x - \frac{1}{2}$. **13.** (a) $y = -\frac{3}{4}x + 5$; y-intercept $(0, 5)$, x-intercept $\left(\frac{20}{3}, 0\right)$. (b) $y = 5x + 8$; y-intercept $(0, 8)$, x-intercept $\left(-\frac{8}{5}, 0\right)$. **14.** (a) $-5 = m(2) - 11$, $2m = 6$, $m = 3$. (b) $4 = -\frac{2}{3}a + 10$, $-6 = -\frac{2}{3}a$, $a = 9$. **15.** (a) $y = -\frac{2}{5}x + 3$ (b) $y = -\frac{2}{5}x + 6$ (c) $m = \frac{5}{2}$: $y = \frac{5}{2}x - 26$. **16.** A. **17.** 5.

Lesson 6 — Extra Practice

Writing the Equation of a Line Given Two Points

Part A — Slope First, Then Substitute

1. Find the slope of the line through each pair of points. Keep every answer exact — several of these do not divide evenly.
(a) $(3, -2)$ and $(7, 10)$ (b) $(-5, 8)$ and $(1, -4)$ (c) $(-6, -1)$ and $(4, 4)$

(d) $(2, 9)$ and $(8, 5)$ (e) $(-7, 3)$ and $(5, -6)$ (f) $(-4, -7)$ and $(6, 8)$

2. Determine the equation, in the form $y = mx + b$, of the line through each pair of points. Every pair has one point in a different quadrant from the other, so substitute each coordinate *with* its sign, in brackets.
(a) $(-3, 11)$ and $(2, -4)$ (b) $(-5, -14)$ and $(3, 10)$

(c) $(4, -9)$ and $(-2, 3)$ (d) $(3, -14)$ and $(-5, 18)$

3. Determine the equation, in the form $y = mx + b$, of the line through each pair of points. The slopes here are fractions — reduce them, then keep them exact right through the substitution.
(a) $(-8, -1)$ and $(4, 8)$ (b) $(-9, 7)$ and $(6, -3)$

(c) $(-10, -9)$ and $(5, 3)$ (d) $(-12, 10)$ and $(8, -5)$

4. Determine the equation of the line through each pair of points. This time *both* m and b come out as fractions. Do not round.
(a) $(-3, 4)$ and $(5, -1)$ (b) $(-6, -5)$ and $(4, 2)$

Part B — Shortcuts, Special Cases, Graphs and Tables

5. In each pair below, one point already sits on the y-axis.

(a) $(0, -7)$ and $(6, 5)$

(b) $(0, 4)$ and $(-10, 9)$

(c) $(0, -3)$ and $(8, -9)$

(d) A point of the form $(0, k)$ lets you skip an entire step of the method. Name the step you can skip and explain, in one sentence, why skipping it is legitimate.

6. Write the equation of the line through each pair of points. None of these four behaves like a normal question — decide what is special *before* you reach for the slope formula.

(a) $(-8, -4)$ and $(5, -4)$

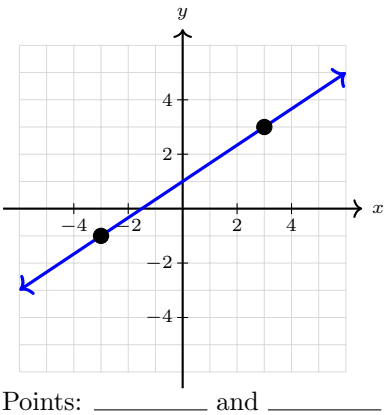
(b) $(-2, 6)$ and $(-2, -9)$; explain why this line cannot be written in the form $y = mx + b$.

(c) $(3, 0)$ and $(-7, 0)$. What is this line usually called?

(d) $(0, 5)$ and $(0, -2)$. Both points are on the y-axis — so is the line the y-axis? Give its equation.

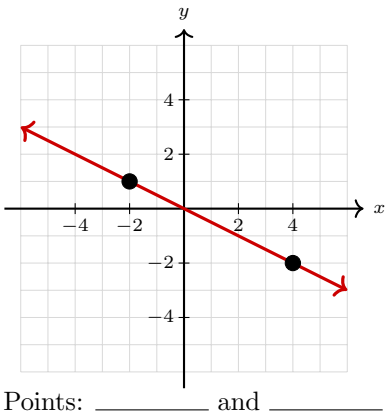
7. Each graph below shows a line with two lattice points marked. Read the two points off the grid, then determine the equation of the line.

(a)



Points: _____ and _____
Slope: _____ b: _____
Equation: _____

(b)



Points: _____ and _____
Slope: _____ b: _____
Equation: _____

8. Each table gives only *two* rows of a linear relation. Determine the equation of the relation, then use it to answer the extra question.

(a)

x	y
-4	13
6	-7

Find y when $x = 12$.

(b)

x	y
-9	-2
3	6

Find y when $x = 15$.

Part C — Applications and Checking

9. A print shop charges a fixed set-up fee plus a constant amount per item. An order of 5 items costs \$32 and an order of 12 items costs \$60. Let C be the cost in dollars for n items.

(a) Determine the equation relating C and n.

(b) Interpret the slope and the C-intercept in the context of this print shop.

(c) Predict the cost of an order of 20 items.

(d) How many items can be ordered for exactly \$100?

10. A pool is being filled at a constant rate. After 4 minutes it holds 130 L; after 10 minutes it holds 205 L. Let V be the volume in litres after t minutes.

(a) Determine the equation relating V and t. The slope is not a whole number — keep it exact. Then state what the slope and the V-intercept mean here.

(b) Predict the volume after 16 minutes, and determine when the pool holds 330 L.

11. **Check both points.** Determine the equation of each line, then verify it by substituting *both* of the original points — not just the one you used — and showing that each gives a true statement.

(a) $(-7, 22)$ and $(5, -14)$

(b) $(-10, -11)$ and $(6, 13)$

Part D — Error Analysis, Working Backwards and Stretch

12. **Error analysis.** In each case, name the mistake, show that the student's equation fails the check at one of the given points, and give the correct equation.

(a) For the line through $(-2, 7)$ and $(4, -5)$, Jaya wrote
$m = \dfrac{-5-7}{-2-4} = \dfrac{-12}{-6} = 2$,
then $7 = 2(-2) + b$, $b = 11$, so $y = 2x + 11$.

(b) For the line through $(3, -4)$ and $(9, 8)$, Omar wrote
$m = \dfrac{9-3}{8-(-4)} = \dfrac{6}{12} = \dfrac{1}{2}$,
then $-4 = \dfrac{1}{2}(3) + b$, $b = -\dfrac{11}{2}$, so $y = \dfrac{1}{2}x - \dfrac{11}{2}$.

13. **Work backwards.** Set up an equation for the unknown in each part and solve it.

(a) A line through $(2, 5)$ has equation $y = 3x + b$. Determine b, then write the equation.

(b) A line through $(1, 4)$ and $(k, 10)$ has slope 2. Determine k.

(c) A line through $(-3, c)$ and $(5, -1)$ has slope $-\dfrac{1}{2}$. Determine c.

(d) A line through $(-6, 2)$ and $(2, t)$ has slope $\dfrac{3}{4}$. Determine t.

14. **Work backwards, continued.**

(a) A line passes through $(-5, -8)$ and $(7, 1)$. Determine its equation and *both* of its intercepts. Give exact coordinates.

(b) A line passes through $(-2, 5)$ and $(6, q)$, and its y-intercept is 3. Determine the value of q.

15. **Stretch — three points.** Consider $A(-5, -7)$, $B(1, 5)$ and $C(4, 11)$.

(a) Show that A, B and C are collinear by comparing slopes.

(b) Write the single equation, in the form $y = mx + b$, of the line containing all three.

(c) Determine p so that $(p, -1)$ also lies on this line.

16. **Stretch — midpoints.** Segment PQ joins $P(-6,1)$ to $Q(2,7)$. Segment RS joins $R(0,-5)$ to $S(8,-1)$.
(a) Determine the midpoint of PQ and the midpoint of RS.

(b) Determine the equation of the line through those two midpoints. Leave m and b exact.

17. *(MC)* The equation of the line through $(-6,11)$ and $(2,-5)$ is
A. $y=-2x+23$ B. $y=2x+23$
C. $y=-2x-1$ D. $y=-\dfrac{1}{2}x+8$

18. *(NR)* The line through $(-4,-9)$ and $(6,11)$ is written as $y=mx+b$. The value of $m-b$ is _____.

Answer Key. 1. (a) 3 (b) -2 (c) $\frac{1}{2}$ (d) $-\frac{2}{3}$ (e) $-\frac{3}{4}$ (f) $\frac{3}{2}$. **2.** (a) $y=-3x+2$ (b) $y=3x+1$ (c) $y=-2x-1$ (d) $y=-4x-2$. **3.** (a) $y=\frac{3}{4}x+5$ (b) $y=-\frac{2}{3}x+1$ (c) $y=\frac{4}{5}x-1$ (d) $y=-\frac{3}{4}x+1$. **4.** (a) $m=-\frac{5}{8}$, $4=\frac{15}{8}+b$, $b=\frac{17}{8}$: $y=-\frac{5}{8}x+\frac{17}{8}$ (b) $m=\frac{7}{10}$, $-5=-\frac{21}{5}+b$, $b=-\frac{4}{5}$: $y=\frac{7}{10}x-\frac{4}{5}$. **5.** (a) $y=2x-7$ (b) $y=-\frac{1}{2}x+4$ (c) $y=-\frac{3}{4}x-3$ (d) Skip the substitute-and-solve step: a point $(0,k)$ is the y-intercept, so $b=k$ can be read straight off and only the slope still has to be computed. **6.** (a) $y=-4$ (both y-values are -4, so $m=0$) (b) $x=-2$; the run is $-2-(-2)=0$, so the slope formula divides by 0 and is undefined — with no value of m there is no $y=mx+b$ form (c) $y=0$; this is the x-axis (d) Yes — the line is the y-axis, a vertical line with equation $x=0$ (note that $x=0$ is *not* of the form $y=mx+b$). **7.** (a) $(-3,-1)$ and $(3,3)$; $m=\frac{4}{6}=\frac{2}{3}$, $-1=-2+b$, $b=1$: $y=\frac{2}{3}x+1$ (b) $(-2,1)$ and $(4,-2)$; $m=\frac{-3}{6}=-\frac{1}{2}$, $b=0$: $y=-\frac{1}{2}x$. **8.** (a) $m=\frac{-20}{10}=-2$, $b=5$: $y=-2x+5$; at $x=12$, $y=-19$ (b) $m=\frac{8}{12}=\frac{2}{3}$, $b=4$: $y=\frac{2}{3}x+4$; at $x=15$, $y=14$. **9.** (a) $m=\frac{60-32}{12-5}=\frac{28}{7}=4$, $32=4(5)+b$, $b=12$: $C=4n+12$ (b) slope 4: each additional item costs \$4; C-intercept 12: a fixed set-up fee of \$12 charged even for 0 items (c) \$92 (d) $100=4n+12$, $n=22$ items. **10.** (a) $m=\frac{205-130}{10-4}=\frac{75}{6}=\frac{25}{2}$, $130=50+b$, $b=80$: $V=\frac{25}{2}t+80$; the pool fills at $\frac{25}{2}=12.5$ L per minute and already held 80 L when timing began (b) $V=\frac{25}{2}(16)+80=280$ L; $330=\frac{25}{2}t+80$ gives $t=20$ min. **11.** (a) $y=-3x+1$; check $-3(-7)+1=22$ ✓ and $-3(5)+1=-14$ ✓ (b) $y=\frac{3}{2}x+4$; check $\frac{3}{2}(-10)+4=-11$ ✓ and $\frac{3}{2}(6)+4=13$ ✓. **12.** (a) She subtracted the y's as y_2-y_1 but the x's as x_1-x_2; mixing the order flips the sign of the slope. Her equation fails at $(4,-5)$: $2(4)+11=19\neq-5$. Correctly $m=\frac{-5-7}{4-(-2)}=\frac{-12}{6}=-2$, $7=-2(-2)+b$, $b=3$, so $y=-2x+3$. (b) He put the run on top and the rise on the bottom — slope is $\frac{\Delta y}{\Delta x}$, not $\frac{\Delta x}{\Delta y}$. His equation fails at $(9,8)$: $\frac{1}{2}(9)-\frac{11}{2}=-1\neq8$. Correctly $m=\frac{8-(-4)}{9-3}=\frac{12}{6}=2$, $-4=2(3)+b$, $b=-10$, so $y=2x-10$. **13.** (a) $5=3(2)+b$, $b=-1$: $y=3x-1$ (b) $\frac{10-4}{k-1}=2$, $6=2k-2$, $k=4$ (c) $\frac{-1-c}{5-(-3)}=-\frac{1}{2}$, $-1-c=-4$, $c=3$ (d) $\frac{t-2}{2-(-6)}=\frac{3}{4}$, $t-2=6$, $t=8$. **14.** (a) $m=\frac{9}{12}=\frac{3}{4}$, $-8=-\frac{15}{4}+b$, $b=-\frac{17}{4}$: $y=\frac{3}{4}x-\frac{17}{4}$; y-intercept $\left(0,-\frac{17}{4}\right)$, x-intercept $\left(\frac{17}{3},0\right)$ (b) the y-intercept gives a second point $(0,3)$, so $m=\frac{5-3}{-2-0}=-1$ and $y=-x+3$; at $x=6$, $q=-3$. **15.** (a) $m_{AB}=\frac{5-(-7)}{1-(-5)}=\frac{12}{6}=2$, $m_{BC}=\frac{11-5}{4-1}=\frac{6}{3}=2$; equal slopes from a shared point means all three lie on one line (b) $-7=2(-5)+b$, $b=3$: $y=2x+3$ (check B: $2(1)+3=5$ ✓; C: $2(4)+3=11$ ✓) (c) $-1=2p+3$, $p=-2$. **16.** (a) midpoint of PQ is $(-2,4)$; midpoint of RS is $(4,-3)$ (b) $m=\frac{-3-4}{4-(-2)}=-\frac{7}{6}$, $4=\frac{7}{3}+b$, $b=\frac{5}{3}$: $y=-\frac{7}{6}x+\frac{5}{3}$. **17.** C. **18.** 3.

Lesson 7 — Extra Practice

The General Form of a Linear Equation

Part A — Clearing Denominators into General Form

Throughout this set, **proper general form** means $Ax + By + C = 0$ where A, B, and C are integers, no fractions appear anywhere, A is positive, and 0 stands alone on the right side. When the slope or the y-intercept is a fraction, multiply *every* term by the common denominator first — that step is the whole job.

1. Convert each equation to proper general form.

(a) $y = \frac{2}{3}x - 4$

(b) $y = -\frac{3}{5}x + 2$

(c) $y = \frac{5}{2}x - 7$

(d) $y = -\frac{1}{6}x + 5$

2. Now *both* the slope and the y-intercept are fractions, so the common denominator has to clear two fractions at once. Convert each equation to proper general form.

(a) $y = \frac{3}{4}x - \frac{2}{3}$

(b) $y = -\frac{5}{6}x + \frac{1}{4}$

(c) $y = \frac{2}{5}x + \frac{3}{10}$

(d) $y = -\frac{7}{2}x - \frac{5}{8}$

3. Convert each equation to slope-intercept form. In (a) and (c) the coefficient of y is negative, so watch every sign as you divide; in (b) and (d) the answer keeps a fraction.

(a) $6x - 9y + 18 = 0$

(b) $10x + 4y - 14 = 0$

(c) $7x - 3y - 6 = 0$

(d) $2x - 12y + 9 = 0$

Part B — Reading a Line Straight Out of General Form

4. Decide whether each equation is written in **proper general form**. If it is not, name the rule it breaks and rewrite it correctly.

Equation	**Rule broken (if any)**	**Proper general form**
(a) $2x - y = 7$		
(b) $-5x + 3y + 10 = 0$		
(c) $\frac{1}{2}x + y - 4 = 0$		
(d) $3x = 4y - 9$		
(e) $4x + 5y - 11 = 0$		

5. For each line, state the slope and *both* intercepts without ever converting to slope-intercept form. (For $Ax + By + C = 0$ the slope is $m = -\dfrac{A}{B}$.) Then answer part (c).

(a) $4x - 5y - 20 = 0$

(b) $3x + 8y + 24 = 0$

(c) Explain why the intercepts are quicker to find from general form than from slope-intercept form, while the slope is quicker from slope-intercept form.

6. Two equations describe the **same line** exactly when one is a multiple of the other — every coefficient scaled by the same number. Decide whether each pair describes the same line, and justify your decision.

(a) $2x - 3y + 6 = 0$ and $6x - 9y + 18 = 0$

(b) $4x + 6y - 10 = 0$ and $2x + 3y - 5 = 0$

(c) $3x - y + 2 = 0$ and $6x - 2y + 5 = 0$

7. From $Ax + By + C = 0$ the slope is $m = -\dfrac{A}{B}$, so two lines can be compared without converting either one. Decide whether each pair is parallel, perpendicular, or neither.

(a) $3x - 4y + 8 = 0$ and $6x - 8y - 5 = 0$

(b) $2x + 5y - 10 = 0$ and $5x - 2y + 3 = 0$

(c) $x + 3y - 6 = 0$ and $3x + y - 6 = 0$

(d) $4x + 6y - 9 = 0$ and $6x - 4y + 7 = 0$

8. Order these four lines from **least steep** to **steepest** using only $m = -\dfrac{A}{B}$. (Steepness compares the size of the slope, ignoring its sign.)

(a) $3x - 2y + 7 = 0$ (b) $5x + 4y - 1 = 0$ (c) $x - 3y + 2 = 0$ (d) $7x + 2y + 9 = 0$

Part C — Errors, Working Backwards, and Context

9. Priya converted $y = 5x - 3$ and wrote $5x + y - 3 = 0$. Show that her equation is a *different* line by converting it back to slope-intercept form. Name the single mistake she made, then give the correct general form.

10. Omar converted $y = \frac{3}{4}x + 2$ and wrote $\frac{3}{4}x - y + 2 = 0$. His equation *is* the same line — so why is it still marked wrong? Finish the job properly.

11. **Work backwards.** Each equation below is missing one number. Substitute the information given and solve for it.

(a) Find C so that $3x - 4y + C = 0$ passes through $(2, -1)$.

(b) Find A so that $Ax + 3y - 12 = 0$ has an x-intercept of 4.

(c) Find B so that $5x + By - 20 = 0$ has slope $-\frac{5}{2}$.

12. Find C so that $2x + 5y + C = 0$ has a y-intercept of 3, then state the x-intercept of the resulting line.

13. **Budget line.** A student has \$40 to spend at a concession stand. Slices of pizza cost \$5 and drinks cost \$2. If x is the number of slices and y is the number of drinks, the possible purchases satisfy $5x + 2y = 40$.

(a) Rewrite the relation in proper general form.

(b) Find both intercepts and explain what each one means in this situation.

(c) State the slope and explain what it means about trading drinks for pizza.

14. **Two jobs.** Rani is paid \$12 per lawn mowed and \$8 per hour of babysitting, and she wants to earn exactly \$96 this week. With x lawns and y hours, this is $12x + 8y - 96 = 0$.

(a) Find both intercepts and say what each one means.

(b) State the slope, explain what it means, and then explain why only some points on this line make sense for Rani.

15. A line in general form is $6x + ky - 18 = 0$.
(a) Find k if the y-intercept is -2. (b) Find k if the slope is -3. (c) What happens to the graph when $k = 0$? Describe the line.

16. **Stretch.** Consider the lines $2x + ky - 6 = 0$ and $3x - y + 1 = 0$.
(a) Find the value of k that makes the two lines **perpendicular**. Verify your answer by multiplying the two slopes. (b) Find the value of k that makes them **parallel**. Your k will not be an integer — multiply the first equation through to show the line can still be written in proper general form.

17. **Stretch.** A line in proper general form is $Ax + By + C = 0$ with A, B, and C all non-zero. Write the x-intercept and the y-intercept in terms of A, B, and C, then use those two expressions to explain why the slope of the line equals $-\dfrac{A}{B}$.

Answer Key. 1. (a) $2x - 3y - 12 = 0$ (b) $3x + 5y - 10 = 0$ (c) $5x - 2y - 14 = 0$ (d) $x + 6y - 30 = 0$. **2.** (a) LCD 12: $9x - 12y - 8 = 0$ (b) LCD 12: $10x + 12y - 3 = 0$ (c) LCD 10: $4x - 10y + 3 = 0$ (d) LCD 8: $28x + 8y + 5 = 0$. **3.** (a) $y = \frac{2}{3}x + 2$ (b) $y = -\frac{5}{2}x + \frac{7}{2}$ (c) $y = \frac{7}{3}x - 2$ (d) $y = \frac{1}{6}x + \frac{3}{4}$. **4.** (a) right side is not 0; $2x - y - 7 = 0$ (b) A is negative; multiply by -1: $5x - 3y - 10 = 0$ (c) a fraction is present; multiply by 2: $x + 2y - 8 = 0$ (d) terms sit on both sides (so the right side isn't 0); $3x - 4y + 9 = 0$ (e) no rule broken — already proper. **5.** (a) $m = \frac{4}{5}$, x-int $(5, 0)$, y-int $(0, -4)$ (b) $m = -\frac{3}{8}$, x-int $(-8, 0)$, y-int $(0, -3)$. (c) In general form both variables are still in the equation, so substituting $y = 0$ or $x = 0$ deletes a whole term and leaves a one-step equation — no rearranging at all. Slope-intercept form, on the other hand, already displays m (and b) with no work, but finding the x-intercept from it still requires solving $0 = mx + b$. **6.** (a) Same line: every coefficient of the second is 3 times the first $(2, -3, 6$ versus $6, -9, 18)$; both give $y = \frac{2}{3}x + 2$. (b) Same line: the first is 2 times the second; both give $y = -\frac{2}{3}x + \frac{5}{3}$. (c) *Not* the same line: $6 = 2(3)$ and $-2 = 2(-1)$, but $5 \neq 2(2)$. Both have slope 3, so they are parallel but distinct ($y = 3x + 2$ versus $y = 3x + \frac{5}{2}$). **7.** (a) $m = \frac{3}{4}$ and $m = \frac{3}{4}$: parallel (and distinct). (b) $m = -\frac{2}{5}$ and $m = \frac{5}{2}$, product -1: perpendicular. (c) $m = -\frac{1}{3}$ and $m = -3$, product 1: neither (a common trap — the reciprocal was flipped but the sign was not changed). (d) $m = -\frac{2}{3}$ and $m = \frac{3}{2}$, product -1: perpendicular. **8.** Slopes: (a) $\frac{3}{2}$, (b) $-\frac{5}{4}$, (c) $\frac{1}{3}$, (d) $-\frac{7}{2}$. Sizes $\frac{1}{3} < \frac{5}{4} < \frac{3}{2} < \frac{7}{2}$, so the order is (c), (b), (a), (d). **9.** $5x + y - 3 = 0$ gives $y = -5x + 3$, not $y = 5x - 3$. She moved $5x$ across without changing its sign. Correct: $y = 5x - 3 \Rightarrow -5x + y + 3 = 0 \Rightarrow 5x - y - 3 = 0$. **10.** General form requires *integer* coefficients, and $\frac{3}{4}$ is a fraction. Multiply every term by 4: $3x - 4y + 8 = 0$. **11.** (a) $3(2) - 4(-1) + C = 0 \Rightarrow 6 + 4 + C = 0 \Rightarrow C = -10$ (b) $4A + 3(0) - 12 = 0 \Rightarrow A = 3$ (c) $-\frac{5}{B} = -\frac{5}{2} \Rightarrow B = 2$. **12.** $2(0) + 5(3) + C = 0 \Rightarrow C = -15$, so $2x + 5y - 15 = 0$; its x-intercept is $\left(\frac{15}{2}, 0\right)$. **13.** (a) $5x + 2y - 40 = 0$. (b) x-int $(8, 0)$: spending everything on pizza buys 8 slices and 0 drinks; y-int $(0, 20)$: spending everything on drinks buys 20 drinks and 0 slices. (c) $m = -\frac{5}{2}$: each extra slice of pizza costs her $\frac{5}{2}$ drinks — equivalently, 2 more slices means 5 fewer drinks. **14.** (a) x-int $(8, 0)$: 8 lawns and no babysitting; y-int $(0, 12)$: 12 hours of babysitting and no lawns. (b) $m = -\frac{3}{2}$: every extra lawn replaces $\frac{3}{2}$ hours of babysitting. Only points with whole-number (or at least non-negative) coordinates make sense, since she cannot mow a negative number of lawns. **15.** (a) $6(0) + k(-2) - 18 = 0 \Rightarrow -2k = 18 \Rightarrow k = -9$ (b) $-\frac{6}{k} = -3 \Rightarrow k = 2$ (c) With $k = 0$ the equation is $6x - 18 = 0$, i.e. $x = 3$: a *vertical* line, slope undefined, with no y-intercept. **16.** (a) $3x - y + 1 = 0$ has slope 3, so the other must have slope $-\frac{1}{3}$: $-\frac{2}{k} = -\frac{1}{3} \Rightarrow k = 6$. Check: $\left(-\frac{2}{6}\right)(3) = -1$. (b) Parallel needs $-\frac{2}{k} = 3 \Rightarrow k = -\frac{2}{3}$. Then $2x - \frac{2}{3}y - 6 = 0$; multiplying by 3 gives $6x - 2y - 18 = 0$, or $3x - y - 9 = 0$ — proper general form, slope 3 as required. **17.** Setting $y = 0$ gives $Ax + C = 0$, so the x-intercept is $\left(-\frac{C}{A}, 0\right)$; setting $x = 0$ gives $By + C = 0$, so the y-intercept is $\left(0, -\frac{C}{B}\right)$. Using those two points, $m = \dfrac{-\frac{C}{B} - 0}{0 - \left(-\frac{C}{A}\right)} = \dfrac{-\frac{C}{B}}{\frac{C}{A}} = -\dfrac{C}{B} \cdot \dfrac{A}{C} = -\dfrac{A}{B}$.

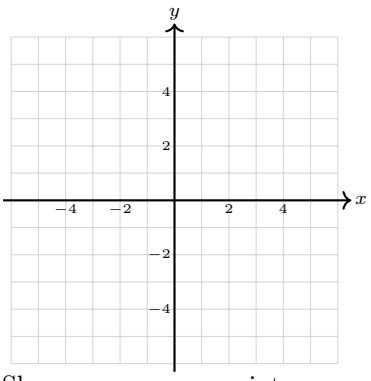
Lesson 8 — Extra Practice

Graphing Horizontal Lines, Vertical Lines, and Any Form

Part A — Horizontal and Vertical Lines on a Grid

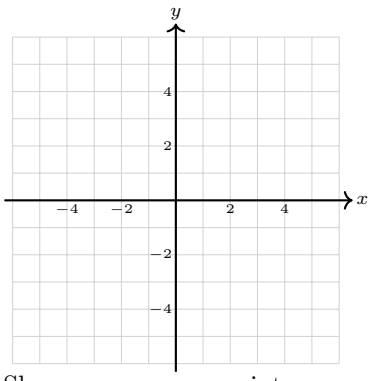
1. Graph each line on its grid, then fill in all three blanks. If a line has no intercept of the kind asked for, write *none* and say why.

(a) $y = -4$



Slope: _____ x-int: _____ y-int: _____

(b) $x = 5$

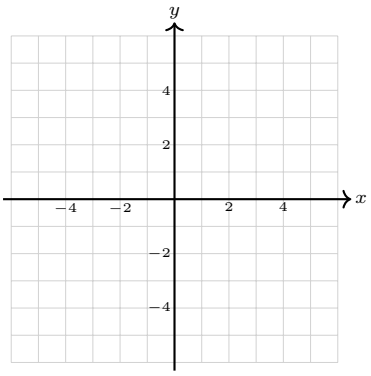


Slope: _____ x-int: _____ y-int: _____

2. Here you are given a point instead of an equation. Write the equation of the line described, then graph it.

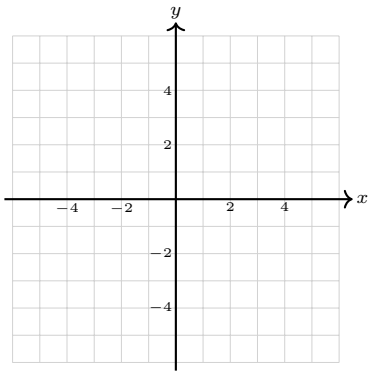
(a) The **horizontal** line through $(2, -5)$.

Equation: _____



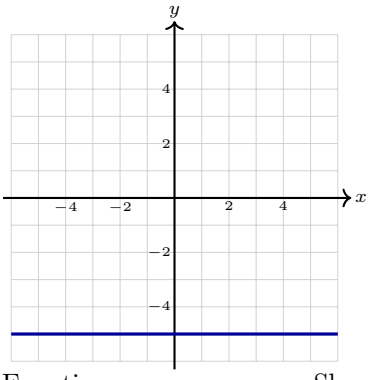
(b) The **vertical** line through $(-4, 3)$.

Equation: _____



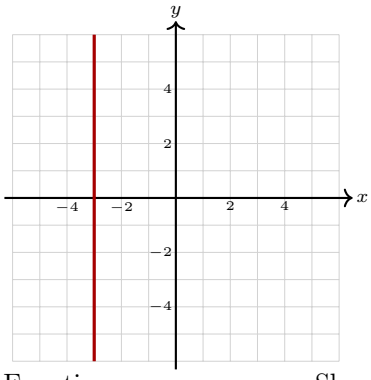
3. Each grid already shows a line. Write its equation and state its slope.

(a)



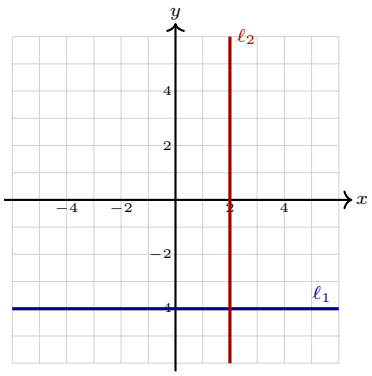
Equation: _____ Slope: _____

(b)



Equation: _____ Slope: _____

4. The grid below shows two lines.



(a) Write the equation of ℓ_1 and the equation of ℓ_2.
(b) State the slope of each line.
(c) Give the coordinates of the point where they cross, and explain how you can read that point off the two equations without looking at the graph at all.

5. All four lines below pass through the point $(-3, 6)$. Write the equation of each one.

(a) parallel to the x-axis

(b) parallel to the y-axis

(c) perpendicular to the x-axis

(d) perpendicular to the y-axis

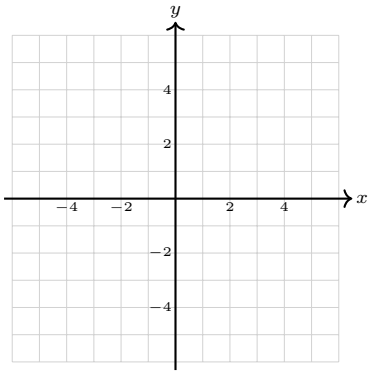
Part B — One Strategy for Any Form

6. Before graphing anything, decide *how* you would graph each line and say why that route is the cheapest. Choose from: **constant** ($y = k$ or $x = k$, no work), **slope-intercept** (plot b, then count the slope), **two-intercept** (set $y = 0$, then set $x = 0$), or **two points** (plot and join).

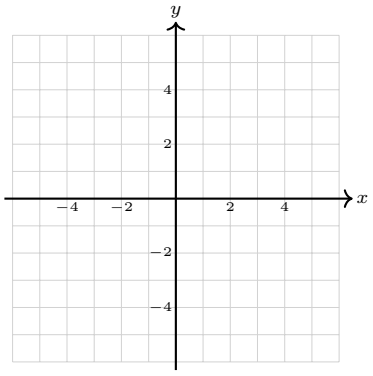
Line	**Method**	**Why that one?**
(a) $y = \frac{3}{4}x - 2$		
(b) $3x + 4y - 12 = 0$		
(c) $x = -5$		
(d) through $(-5, 2)$ and $(1, -4)$		

7. Graph each line, and name the method you used. These two are deliberately mismatched: one has an ugly x-intercept, the other has tidy intercepts.

(a) $y = -\frac{2}{3}x + 3$
Method: _____



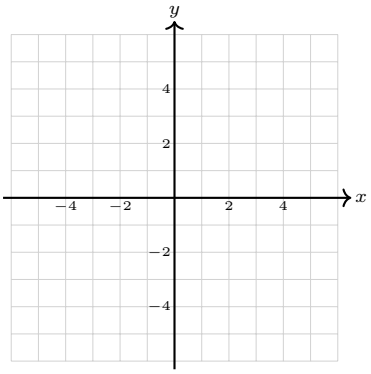
(b) $3x + 5y + 15 = 0$
Method: _____



8. Graph each line using the two-intercept method. Write both intercepts before you draw.

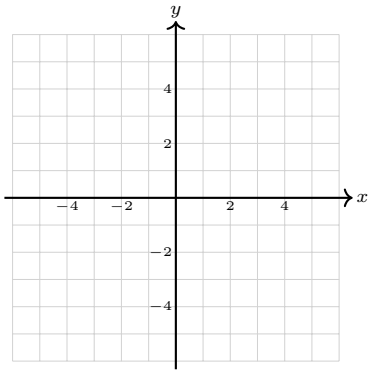
(a) $2x - 3y - 6 = 0$

x-int: _____ y-int: _____



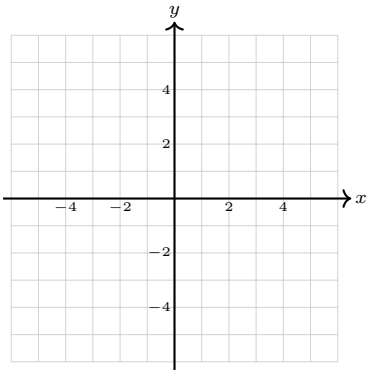
(b) $5x + 2y + 10 = 0$

x-int: _____ y-int: _____

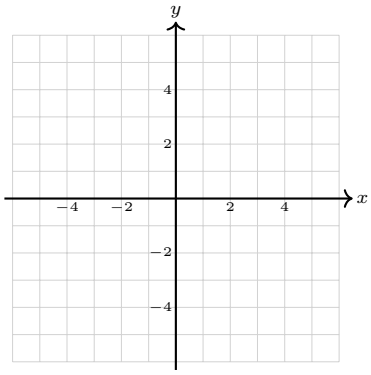


9. In (a) you are given two points and no equation; in (b) you are given three lines to draw on one grid.

(a) Graph the line through $(-4, 5)$ and $(2, -1)$, then write its equation in slope-intercept form *and* in general form.



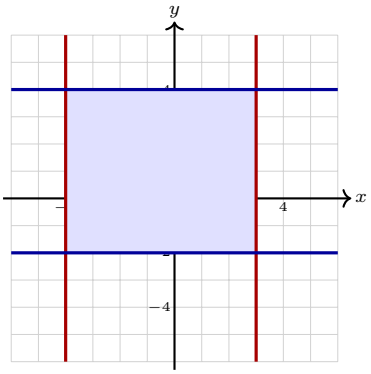
(b) On one grid, draw $x = -3$, $y = 2$, and $y = -x + 1$. State the three points where the lines cross one another.



10. Which of the lines $y = 4$, $x = 4$, $y = -\frac{1}{2}x + 4$ and $2x + y - 4 = 0$ share a point with *all* of the others? Find every crossing point you can and explain what makes the answer come out the way it does.

Part C — Figures, Errors, Reasoning, and Working Backwards

11. The rectangle below is bounded by two horizontal lines and two vertical lines.

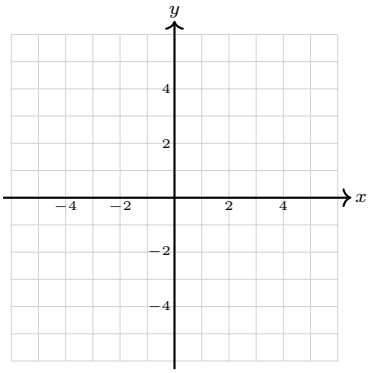


(a) Write the equations of all four lines.

(b) Give the coordinates of all four corners of the rectangle.

(c) Find the area and the perimeter of the rectangle, in square units and units.

12. Asked for the equation of the **vertical** line through $(3, 0)$, Devon wrote $y = 3$. Graph Devon's line and the correct line on the same axes to show they are different, then explain the swap he made in one sentence.



13. Maya says the line $x = 4$ has slope 0 "because y never changes." Two things are wrong with that sentence. Name both, and give the correct slope.

14. Using $m = \dfrac{\text{rise}}{\text{run}}$ and two points of your own choosing on $x = -2$, explain carefully why the slope of a vertical line is *undefined* rather than 0. Then explain why $x = -2$ has an x-intercept but no y-intercept.

15. Use the vertical line test to explain why $x = a$ is **not** a function, but $y = b$ **is**. State the domain and range of $x = -2$ and of $y = 5$ to support your answer.

16. **Work backwards.** Write the equation of each line described.

(a) slope 0, passing through $(-2, 5)$

(b) undefined slope, passing through $(-2, 5)$

(c) slope 0 with y-intercept -3

(d) vertical, with x-intercept 6

(e) vertical, passing through the origin

(f) horizontal, 4 units below the x-axis

17. A line is parallel to $x = -1$ and passes through $(7, -2)$; a second line is perpendicular to $x = -1$ and passes through the same point. Write both equations, state both slopes, and explain why neither answer needs any calculation.

18. **Stretch.** The three lines $x = -2$, $y = -3$, and $x + 2y - 6 = 0$ enclose a triangle.
(a) Find the three vertices by solving the lines in pairs.
(b) Explain why the triangle must have a right angle, and say at which vertex.
(c) Find the area of the triangle.

Answer Key. 1. (a) Horizontal line through $(0, -4)$; slope 0; y-int -4; no x-intercept — it runs parallel to the x-axis and never meets it. (b) Vertical line through $(5, 0)$; slope undefined; x-int 5; no y-intercept. **2.** (a) $y = -5$ (the x-coordinate 2 is irrelevant) (b) $x = -4$ (the y-coordinate 3 is irrelevant). **3.** (a) $y = -5$, slope 0 (b) $x = -3$, slope undefined. **4.** (a) ℓ_1: $y = -4$; ℓ_2: $x = 2$. (b) ℓ_1 slope 0; ℓ_2 slope undefined. (c) $(2, -4)$. Each equation pins one coordinate, so together they name exactly one point — x must be 2 and y must be -4. **5.** (a) $y = 6$ (b) $x = -3$ (c) $x = -3$ (d) $y = 6$. Parallel to an axis and perpendicular to the *other* axis describe the same line. **6.** (a) slope-intercept — $b = -2$ is given and the slope $\frac{3}{4}$ counts out to a lattice point (right 4, up 3). (b) two-intercept — $y = 0$ gives $(4, 0)$ and $x = 0$ gives $(0, 3)$, both whole numbers, with no rearranging. (c) constant — draw the vertical line through -5; no calculation at all. (d) two points — both points are already given, so just plot and join. **7.** (a) Slope-intercept. Plot $(0, 3)$ and count down 2, right 3 to $(3, 1)$. (Its x-intercept is $\frac{9}{2}$, which is why the two-intercept method would be awkward here.) (b) Two-intercept: x-int $(-5, 0)$, y-int $(0, -3)$; join them. Equivalent to $y = -\frac{3}{5}x - 3$. **8.** (a) x-int $(3, 0)$, y-int $(0, -2)$; the line is $y = \frac{2}{3}x - 2$. (b) x-int $(-2, 0)$, y-int $(0, -5)$; the line is $y = -\frac{5}{2}x - 5$. **9.** (a) $m = \dfrac{-1-5}{2-(-4)} = \dfrac{-6}{6} = -1$, so $y = -x + 1$; in general form, $x + y - 1 = 0$. (b) $x = -3$ and $y = 2$ cross at $(-3, 2)$; $x = -3$ and $y = -x + 1$ cross at $(-3, 4)$; $y = 2$ and $y = -x + 1$ cross at $(-1, 2)$. **10.** $y = 4$ and $x = 4$ cross at $(4, 4)$; $y = 4$ and $y = -\frac{1}{2}x + 4$ cross at $(0, 4)$; $y = 4$ and $2x + y - 4 = 0$ cross at $(0, 4)$; $x = 4$ and $y = -\frac{1}{2}x + 4$ cross at $(4, 2)$; $x = 4$ and $2x + y - 4 = 0$ cross at $(4, -4)$; $y = -\frac{1}{2}x + 4$ and $2x + y - 4 = 0$ cross at $(0, 4)$. Every line meets every other line, so all four qualify — no two of them are parallel (slopes 0, undefined, $-\frac{1}{2}$ and -2 are all different), and two non-parallel lines always meet exactly once. Three of them happen to share the single point $(0, 4)$. **11.** (a) Vertical: $x = -4$ and $x = 3$. Horizontal: $y = -2$ and $y = 4$. (b) $(-4, -2)$, $(3, -2)$, $(3, 4)$, $(-4, 4)$. (c) Width $3 - (-4) = 7$, height $4 - (-2) = 6$; area = 42 square units, perimeter $= 2(7 + 6) = 26$ units. **12.** $y = 3$ is horizontal through $(0, 3)$; the correct line is $x = 3$, vertical through $(3, 0)$. Devon used the number 3 but attached it to the wrong variable — in $x = k$ the fixed coordinate is x, so the point $(3, 0)$ gives $x = 3$. **13.** First, on $x = 4$ it is x that never changes, not y — y takes every value. Second, even her reason would not give slope 0: rise over run here is (something)$\div 0$, which is undefined, not zero. The slope of $x = 4$ is **undefined**. **14.** Take $(-2, 1)$ and $(-2, 5)$ on $x = -2$: rise $= 5 - 1 = 4$, run $= -2 - (-2) = 0$, so $m = \frac{4}{0}$, and division by zero has no value — undefined. Slope 0 would mean $\frac{0}{\text{run}}$, which is the opposite situation (no rise, plenty of run). The line crosses the x-axis at $(-2, 0)$, giving an x-intercept of -2; but it is parallel to the y-axis and never touches it, so there is no y-intercept. **15.** A vertical line drawn at $x = a$ meets the graph of $x = a$ at infinitely many points, so the vertical line test fails and $x = a$ is not a function — one input $x = a$ is paired with every y. Any vertical line meets $y = b$ exactly once, so $y = b$ is a function. Domain of $x = -2$: $\{-2\}$; range: all real numbers. Domain of $y = 5$: all real numbers; range: $\{5\}$. **16.** (a) $y = 5$ (b) $x = -2$ (c) $y = -3$ (d) $x = 6$ (e) $x = 0$ (the y-axis) (f) $y = -4$. **17.** Parallel to $x = -1$ means vertical, so $x = 7$, slope undefined. Perpendicular to $x = -1$ means horizontal, so $y = -2$, slope 0. Neither needs calculation: the first copies the x-coordinate of the point, the second copies its y-coordinate — the whole question is deciding which coordinate to keep. **18.** (a) $x = -2$ with $y = -3$: $(-2, -3)$. $x = -2$ with $x + 2y - 6 = 0$: $-2 + 2y - 6 = 0$, so $y = 4$ and the vertex is $(-2, 4)$. $y = -3$ with $x + 2y - 6 = 0$: $x - 6 - 6 = 0$, so $x = 12$ and the vertex is $(12, -3)$. (b) $x = -2$ is vertical and $y = -3$ is horizontal, and those two are always perpendicular, so the right angle is at their intersection, $(-2, -3)$. (c) The legs are the vertical segment of length $4 - (-3) = 7$ and the horizontal segment of length $12 - (-2) = 14$, so the area is $\frac{1}{2}(7)(14) = 49$ square units.

Lesson 9 — Extra Practice

Slope in Context: Interpreting Rates of Change

Part A — Reading the Model: Slope, Intercept, and Units

1. For each model, state the slope, give its **units**, and write one full sentence explaining what it means in the situation. Say whether the quantity is rising or falling.

(a) A phone plan costs $C = 0.09n + 38$, where C is the monthly bill in dollars and n is the number of long-distance minutes used.

(b) A tank holds $V = -24t + 960$, where V is in litres and t is the number of minutes since the valve was opened.

(c) A taxi fare is $F = 1.85k + 3.60$, where F is in dollars and k is the distance travelled in kilometres.

(d) Air temperature is $T = -6.2a + 18$, where T is in °C and a is the altitude in kilometres above sea level.

2. For each model, state the y-intercept **with its units** and write one full sentence explaining what it represents. Remember that an intercept never carries a "per" unit.

(a) A savings balance is $S = 45w + 380$, where S is in dollars and w is the number of weeks of saving.

(b) A coffee card's balance is $B = -3.25c + 91$, where B is in dollars and c is the number of coffees bought. Also state the units of the slope.

(c) Snow depth is $D = 1.4h + 12$, where D is in centimetres and h is the number of hours the storm has lasted.

(d) A courier charges $C = 2.75p + 9.50$, where C is in dollars and p is the number of parcels delivered.

3. A text-message plan costs $C = 0.05n + 24.50$, where C is the monthly bill in dollars and n is the number of texts sent.

(a) Interpret the slope in a full sentence with units, then interpret the C-intercept in a full sentence with units.

(b) Find the bill in a month with 320 texts. Then find the number of texts that would produce a bill of exactly \$49.50.

4. A graph is drawn for each situation below. Without any equation, state **the units of the slope** of the line. Write them in the form "units of the vertical axis per unit of the horizontal axis."

(a) vertical: volume (L); horizontal: time (s)

(b) vertical: cost ($); horizontal: mass (kg)

(c) vertical: distance (m); horizontal: time (s)

(d) vertical: time (min); horizontal: distance (km)

Part B — Building a Model and Using It Both Ways

5. A scuba tank holds 2400 L of compressed air. A diver breathes it at a steady 40 L per minute.

(a) Write an equation for the air A (litres) remaining after t minutes.

(b) How much air remains after 18 minutes?

(c) Divers surface when 600 L are left. After how many minutes is that?

(d) When would the tank be completely empty?

6. A van rental costs a flat $54, plus $0.25 for every kilometre driven.

(a) Write an equation for the total cost C (dollars) after d kilometres.

(b) Find the cost of a 250 km trip.

(c) Marisol can spend at most $180. What is the greatest distance she can drive?

7. A hot-air balloon descends at a constant rate. Four minutes after the pilot began the descent the balloon was at 1120 m; six minutes later, at 10 minutes, it was at 700 m.

(a) Determine the rate of change of the balloon's altitude, with units.

(b) Write an equation for the altitude A (metres) after t minutes of descent.

(c) Interpret the A-intercept in this situation.

(d) At what time does the balloon touch down?

8. On a clear day the air cools steadily as you climb. At an altitude of 500 m the temperature was 14.5°C; at 2000 m it was 5.5°C.

(a) Determine the rate of change in °C per metre, then restate the same rate in °C per kilometre. Explain why the two numbers look so different but mean the same thing.

(b) Write an equation for the temperature T (°C) at an altitude of h metres, then interpret the T-intercept.

(c) Predict the temperature at 3200 m.

(d) At what altitude does the temperature reach -5°C?

Part C — Changing the Units Inside the Rate

A rate keeps its meaning when its units change, but the *number* changes with it. A pump moving 300 L per hour moves 5 L each minute, because one hour is 60 minutes: $\dfrac{300\text{ L}}{1\text{ h}} = \dfrac{300\text{ L}}{60\text{ min}} = 5\text{ L/min}$. Always convert the rate **before** putting it into $y = mx + b$, so that the slope and the variable x use the same unit of time.

9. A pump fills a trough at a constant 720 litres per hour. The trough already holds 150 L when the pump is switched on.

(a) Express the pumping rate in litres per minute.

(b) Write an equation for the volume V (litres) after m **minutes**.

(c) How much water is in the trough after 25 minutes?

(d) When does the trough hold 870 L?

10. A candle burns down at a steady 0.9 cm per hour. It is 18 cm tall when lit.

(a) Express the burn rate in cm per minute and in mm per hour.

(b) Write an equation for the height H (cm) after t **hours**.

(c) How tall is the candle after 150 minutes?

(d) How long until the candle burns out?

11. A train travels at a constant 90 km/h. It passes a marker that is 12 km from the station, moving away from the station.

(a) Express the speed in km per minute and in metres per second.

(b) Write an equation for the distance D (km) from the station after t **minutes**.

(c) How far from the station is the train after 40 minutes?

(d) When is the train 150 km from the station?

Part D — Comparing Two Rates

12. Two phone plans charge for long-distance minutes as follows, with C in dollars and n the number of minutes:

$$\text{Plan A: } C = 0.11n + 26 \qquad \text{Plan B: } C = 0.19n + 14$$

(a) Interpret both slopes and both intercepts with units, then say in one sentence which plan charges more per minute and which charges more before any calls are made.

(b) Determine the break-even number of minutes, and the bill at that point. Check your answer by substituting into *both* equations.

(c) Which plan is cheaper for 100 minutes? Show both bills.

(d) Which plan is cheaper for 400 minutes? Show both bills, and explain why the winner changed.

13. **Stretch.** Tank X holds 480 L and is draining at 15 L/min. At the same moment, Tank Y holds 300 L and is being filled at 9 L/min.

(a) Write an equation for the volume in each tank after t minutes, and interpret each slope with units.

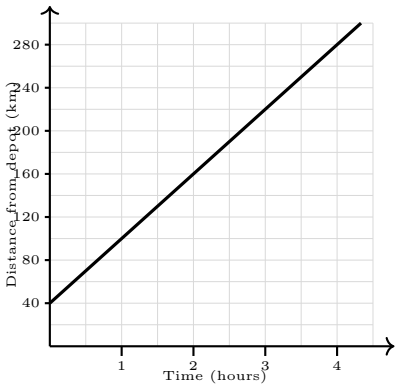
(b) At what time do the two tanks hold the same amount? How much is in each tank then?

(c) Which tank holds more after 5 minutes? After 20 minutes? Support each answer with both volumes.

(d) Tank X eventually runs dry. When? How much is in Tank Y at that instant, and why can Tank X never catch up again?

Part E — Rates from Graphs and Tables

14. The graph shows a truck's distance from a depot during one delivery run. Read the axis labels carefully — the squares are not one unit each.



(a) Determine the rate of change from the graph, with units.

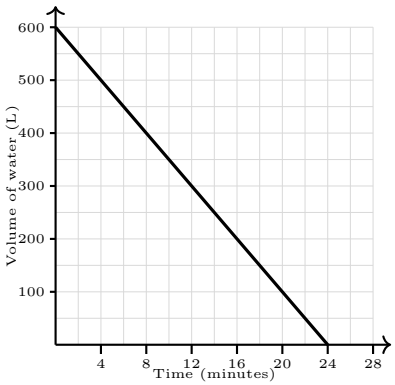
(b) State the y-intercept with units and explain what it means for this truck.

(c) Write the equation of the line using D for distance and t for time.

(d) How far has the truck travelled after 3.5 hours?

(e) The run ends at 400 km. Use the equation — not the graph — to find how long it lasts.

15. The graph shows the water remaining in a reservoir tank as it drains.



(a) Determine the rate of change from the graph, with units, and explain what its sign means.

(b) Write the equation of the line using V for volume and t for time.

(c) How much water is left after 10 minutes? Confirm your answer on the graph.

(d) When is the tank empty? Which feature of the graph gives this directly?

16. A sensor recorded the depth of water in a cattle trough as it was siphoned out.

Time (min)	0	5	10	15	20
Depth (cm)	62	57.5	53	48.5	44

(a) Show that the rate of change is constant, and state it with units.

(b) Write an equation for the depth d (cm) after t minutes, and interpret the d-intercept.

(c) Predict the depth after 35 minutes.

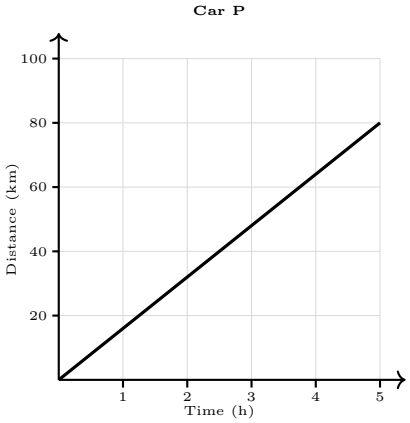
(d) The siphon must be stopped at a depth of 8 cm. When is that?

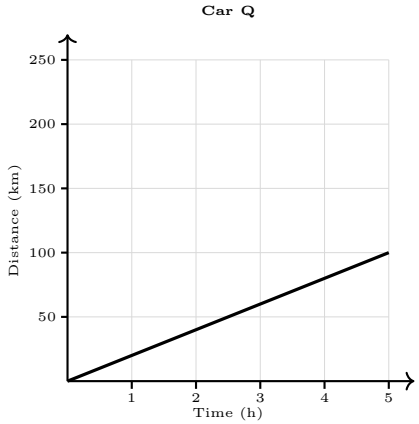
Part F — Error Analysis and Reasoning

17. Asked what the slope of $C = 2.40k + 15$ means (with C in dollars and k in kilograms), Devon wrote: "The slope is 2.40, so the cost goes up by 2.40." Explain what is missing from that answer, then write a complete interpretation.

18. A tank drains steadily. Brooke computed $\dfrac{6 \text{ min} - 2 \text{ min}}{240 \text{ L} - 320 \text{ L}} = -0.05$ and wrote "the tank drains at 0.05 L per minute." Identify **two** errors, then state the correct rate with correct units.

19. **Does a steeper-looking line always mean a faster rate?** The two graphs below show two cars on a highway. Study both scales before you answer.





(a) Which line *looks* steeper on the page?

(b) Compute each car's speed in km/h.

(c) Which car is actually faster? Explain what made the picture misleading, and state the condition under which "steeper means faster" is a safe conclusion.

20. **Stretch.** One summer job pays \$0.42 per minute worked; another pays \$24.60 per hour. Convert both rates to the same units — first to dollars per hour, then to dollars per minute — and say which job pays more. Explain why comparing 0.42 with 24.60 directly tells you nothing.

21. Two students model the same draining tank. Ama writes $V = -18t + 540$ with t in minutes; Ben writes $V = -1080t + 540$ with t in hours. Can both be right? Show the conversion, then state the volume each model predicts at $t = 15$ min and confirm they agree.

Answer Key. 1. (a) $m = 0.09$ \$/min: the bill rises \$0.09 for each extra long-distance minute; rising. (b) $m = -24$ L/min: the tank loses 24 L every minute; falling. (c) $m = 1.85$ \$/km: the fare rises \$1.85 for each additional kilometre; rising. (d) $m = -6.2$ °C/km: the air cools 6.2°C for every kilometre gained in altitude; falling. **2.** (a) $b = \$380$: the amount already saved before any weeks passed. (b) $b = \$91$: the value of the card before any coffee was bought; slope units are dollars per coffee. (c) $b = 12$ cm: the snow already on the ground when the storm began. (d) $b = \$9.50$: the base charge before any parcels are delivered. **3.** (a) $m = \$0.05$ per text — each text adds 5 cents to the bill; $b = \$24.50$ — the monthly charge with no texts sent. (b) $0.05(320) + 24.50 = \$40.50$; $49.50 = 0.05n + 24.50 \Rightarrow 0.05n = 25 \Rightarrow n = 500$ texts. **4.** (a) L/s (b) \$/kg (c) m/s (d) min/km. **5.** (a) $A = -40t + 2400$ (b) 1680 L (c) 45 min (d) 60 min. **6.** (a) $C = 0.25d + 54$ (b) \$116.50 (c) 504 km. **7.** (a) $m = \dfrac{700 - 1120}{10 - 4} = -70$ m/min: the balloon falls 70 m each minute. (b) $A = -70t + 1400$ (c) 1400 m — the altitude when the descent began (d) $t = 20$ min. **8.** (a) $m = \dfrac{5.5 - 14.5}{2000 - 500} = -0.006$ °C/m, which is -6 °C/km; a kilometre is 1000 times a metre, so the number is 1000 times larger while the rate itself is unchanged. (b) $T = -0.006h + 17.5$; the T-intercept 17.5°C is the temperature at sea level ($h = 0$). (c) $T = -0.006(3200) + 17.5 = -1.7$°C (d) $-5 = -0.006h + 17.5 \Rightarrow h = 3750$ m. **9.** (a) $720 \div 60 = 12$ L/min (b) $V = 12m + 150$ (c) 450 L (d) 60 min. **10.** (a) $0.9 \div 60 = 0.015$ cm/min; $0.9 \times 10 = 9$ mm/h (b) $H = -0.9t + 18$ (c) 150 min = 2.5 h, so $H = -0.9(2.5) + 18 = 15.75$ cm (d) $18 \div 0.9 = 20$ h. **11.** (a) $90 \div 60 = 1.5$ km/min; $90\,000 \div 3600 = 25$ m/s (b) $D = 1.5t + 12$ (c) 72 km (d) $150 = 1.5t + 12 \Rightarrow t = 92$ min. **12.** (a) A: \$0.11 per minute and \$26 before any calls; B: \$0.19 per minute and \$14 before any calls — B charges more per minute, A charges more up front. (b) $0.11n + 26 = 0.19n + 14 \Rightarrow 0.08n = 12 \Rightarrow n = 150$ min; both give $C = \$42.50$. (c) 100 min: A = \$37, B = \$33, so B is cheaper. (d) 400 min: A = \$70, B = \$90, so A is cheaper — B's higher per-minute rate eventually outweighs A's larger flat fee. **13.** (a) X: $V = -15t + 480$ (losing 15 L/min); Y: $V = 9t + 300$ (gaining 9 L/min). (b) $-15t + 480 = 9t + 300 \Rightarrow 24t = 180 \Rightarrow t = 7.5$ min, with 367.5 L in each. (c) At 5 min X has 405 L and Y has 345 L, so X holds more; at 20 min X has 180 L and Y has 480 L, so Y holds more. (d) X empties at $t = 32$ min, when Y holds 588 L; X is falling while Y is rising, so once Y passes X the gap only widens. **14.** (a) Rise 240 km over 4 h = 60 km/h. (b) 40 km — the truck was already 40 km from the depot when timing started. (c) $D = 60t + 40$ (d) $D = 60(3.5) + 40 = 250$ km (e) $400 = 60t + 40 \Rightarrow t = 6$ h. **15.** (a) $\dfrac{0 - 600}{24 - 0} = -25$ L/min; the negative sign means the volume is decreasing. (b) $V = -25t + 600$ (c) $V = -25(10) + 600 = 350$ L (d) $t = 24$ min — the t-intercept, where the line meets the horizontal axis. **16.** (a) Each 5-minute step drops the depth by 4.5 cm, so $m = \dfrac{-4.5}{5} = -0.9$ cm/min for every interval. (b) $d = -0.9t + 62$; the d-intercept 62 cm is the depth when the siphon started. (c) $d = -0.9(35) + 62 = 30.5$ cm (d) $8 = -0.9t + 62 \Rightarrow t = 60$ min. **17.** No units and no quantity named — "goes up by 2.40" could mean anything. Correctly: the cost increases by \$2.40 for each additional kilogram, that is, $m = \$2.40$/kg. **18.** (1) She inverted the slope formula, dividing the change in time by the change in volume; (2) her units are therefore min/L, not L/min, so the label does not match the calculation. Correctly $m = \dfrac{240 - 320}{6 - 2} = -20$ L/min: the tank loses 20 L each minute. **19.**

(a) Car P's line looks steeper. (b) P: $80 \div 5 = 16$ km/h; Q: $100 \div 5 = 20$ km/h. (c) Car Q is faster. The vertical scales differ — one square is 20 km on P's graph but 50 km on Q's — so equal-looking rises are not equal distances. Comparing steepness is only safe when both graphs use the same scale on both axes. **20.** \$0.42/min $\times 60 =$ \$25.20/h, and \$24.60/h $\div 60 =$ \$0.41/min. The first job pays more (\$25.20/h versus \$24.60/h, or \$0.42/min versus \$0.41/min). Comparing 0.42 with 24.60 compares dollars-per-minute with dollars-per-hour — different units, so the numbers are not comparable until one is converted. **21.** Both are right. 18 L/min $\times 60$ min/h $= 1080$ L/h, so the two slopes are the same rate in different time units, and the starting volume 540 L is unchanged. At 15 min Ama gets $-18(15) + 540 = 270$ L; 15 min $= 0.25$ h, so Ben gets $-1080(0.25) + 540 = 270$ L — they agree.

Lesson 10 — Extra Practice

Unit Review: Equations of Lines

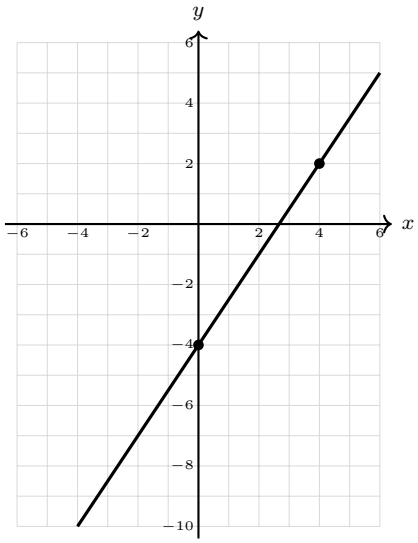
Part A — One Line, Four Ways

1. The four descriptions below all belong to **the same straight line**. Answer each part from its own description alone — that is the point of the question.

Description I — a table of values

x	-4	-2	0	2	4
y	-10	-7	-4	-1	2

Description II — a graph



Description III — two points
The line passes through $(-2, -7)$ and $(6, 5)$.

Description IV — words
The line rises 3 units for every 2 units it runs to the right, and it passes through $(4, 2)$.

(a) Using **Description I** only: state the slope and the y-intercept from the table, then write the equation in the form $y = mx + b$.

(b) Using **Description II** only: read two points off the graph, find the slope, and write the equation in the form $y = mx + b$.

(c) Using **Description III** only: use $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, then solve for b.

(d) Using **Description IV** only: write the slope as a fraction, then solve for b.

(e) Your four answers should be identical. State that one equation, then convert it to general form $Ax + By + C = 0$ with A, B, C integers and A positive. *(You will have to clear a fraction first.)*

(f) Using the *equation* and not the graph: decide whether $(10, 11)$ lies on this line, and find the value of y when $x = -8$.

Part B — Writing Equations From Mixed Information

2. Determine the equation of each line, **and then rewrite your answer in general form**. Two of these four lines will surprise you — read the given information carefully before you reach for $y = mx + b$.

(a) slope $-\frac{3}{4}$, passing through $(8, -1)$

(b) passing through $(-3, 8)$ and $(5, -4)$

(c) passing through $(-5, 6)$ and $(-5, -2)$

(d) passing through $(7, -3)$ and $(-2, -3)$

(e) Exactly one of the four lines above **cannot** be written in the form $y = mx + b$. State which one and explain why in one sentence. Then explain why the other odd-looking line *can* be written that way.

3. Line L has equation $4x - 6y + 9 = 0$.

(a) Convert L to slope-intercept form and state its slope.

(b) Line M is **parallel** to L and passes through $(-6, 5)$. Determine its equation in general form.

(c) Line N is **perpendicular** to L and passes through $(-6, 5)$. Determine its equation in general form.

4. **Work backwards.** Each line below is missing one number. Find it.

(a) The point $(a, 7)$ lies on $5x - 2y - 16 = 0$. Find a.

(b) The point $(-4, k)$ lies on the line through $(2, 5)$ and $(8, 14)$. Find k.

(c) The line $y = mx - 7$ passes through $(3, 2)$. Find m.

(d) The line $3x + By - 12 = 0$ passes through $(2, -3)$. Find B.

5. A table of values is given below. Notice that the x-values do not go up by 1.

x	-6	-3	0	3	6
y	13	9	5	1	-3

(a) Show that the rate of change is constant, then write the equation in the form $y = mx + b$.

(b) Convert your equation to general form.

(c) Predict the value of y when $x = 12$.

(d) Determine the value of x that gives $y = 25$.

Part C — Converting Between Forms

6. Convert each equation to general form $Ax + By + C = 0$, with A, B, C integers and A positive. Clear every fraction first.

(a) $y = -\frac{5}{6}x + \frac{2}{3}$ (b) $y = \frac{7}{4}x - 3$ (c) $y = -\frac{1}{3}x - \frac{5}{2}$ (d) $x = \frac{9}{2}$

7. For each line: convert to slope-intercept form, state the slope and the y-intercept, then give the x-intercept as an ordered pair. Leave every answer as an exact fraction.

(a) $9x - 12y + 8 = 0$ (b) $6x + 15y - 20 = 0$

8. **Same line in different clothes.** Five equations are listed below.

(i) $y = \frac{2}{5}x - 3$ (ii) $2x - 5y - 15 = 0$ (iii) $5y - 2x + 15 = 0$ (iv) $4x - 10y - 30 = 0$ (v) $y = \frac{5}{2}x - 3$

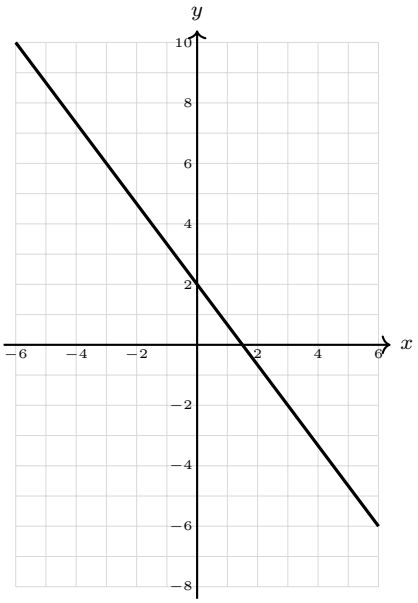
(a) Sort them: which describe the **same** line, and which one is different? Show the work that proves it.

(b) Only one of (ii), (iii) and (iv) is written in **proper** general form. Which one, and what is wrong with each of the other two?

(c) State the slope and y-intercept of the odd line out, and say how you can tell at a glance that it is not the same line.

Part D — Graphs and Lines

9. **From a graph to an equation to a prediction.**



(a) State the coordinates of two points where the line crosses a grid corner exactly.

(b) Determine the slope from those two points.

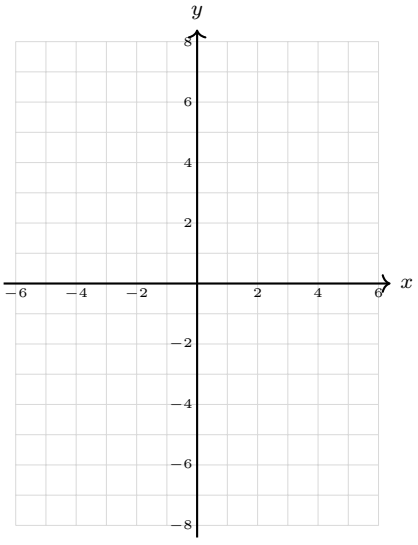
(c) Write the equation in the form $y = mx + b$.

(d) Convert the equation to general form.

(e) The grid stops at $x = 6$. Use the *equation* to find y when $x = 12$, and to find x when $y = -30$.

10. Graph all four lines below on the single grid provided and label each one. Two of them must be rearranged before they can be graphed from a slope and an intercept, and one of them cannot be put in the form $y = mx + b$ at all.

(i) $y = \frac{2}{3}x - 3$ (ii) $x + 2y - 8 = 0$ (iii) $y = -2$ (iv) $x = -5$

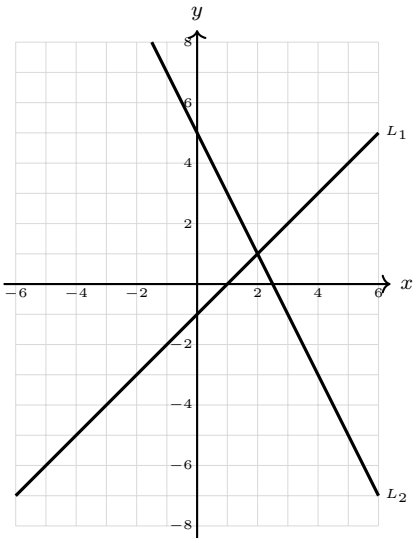


(a) State the slope and y-intercept of each line that has them, showing any rearranging.

(b) Which line has no slope-intercept form, and which line has a slope of 0? Explain the difference.

(c) Two of the four lines meet at a point that sits exactly on a grid corner. Name the two lines and state the point, then verify it by substituting into both equations.

11. Two lines are graphed below.



(a) Read the point of intersection from the graph.

(b) Determine the equation of L_1 in general form.

(c) Determine the equation of L_2 in general form.

(d) Verify your answer to (a) by substituting the point into *both* general-form equations and showing that each gives 0.

Part E — Reasoning and Error Analysis

12. Priya was asked for the equation, in general form, of the line through $(-4, 9)$ and $(2, -3)$. **Exactly one** of her five steps is wrong.

Step 1: $m = \dfrac{-3-9}{2-(-4)} = \dfrac{-12}{6} = -2$
Step 2: $y = -2x + b$
Step 3: $9 = -2(-4) + b$, so $9 = -8 + b$
Step 4: $b = 17$
Step 5: $y = -2x + 17$, so $2x + y - 17 = 0$

(a) Identify the first step that is wrong and explain the error in one sentence.

(b) Redo the solution correctly from that step onward and give the correct general form.

(c) Check your answer by substituting *both* original points into it.

13. Malik converted $y = -\frac{3}{5}x + 2$ to general form as follows. Again, exactly one step is wrong.

Step 1: multiply every term by 5: $5y = -3x + 2$
Step 2: move the x-term across: $3x + 5y - 2 = 0$
Step 3: $A = 3$ is positive, so this is proper general form.

(a) Identify the wrong step and explain the error.

(b) Give the correct general form.

(c) The point $(0, 2)$ must satisfy the correct equation. Substitute it into Malik's equation and into yours to show which one is right.

14. Decide whether each statement is **true** or **false**. If it is false, correct it; if it is true, give a brief reason or an example.

(a) Every straight line can be written in the form $y = mx + b$.

(b) Every straight line can be written in the form $Ax + By + C = 0$.

(c) If two lines have the same y-intercept, they must be parallel.

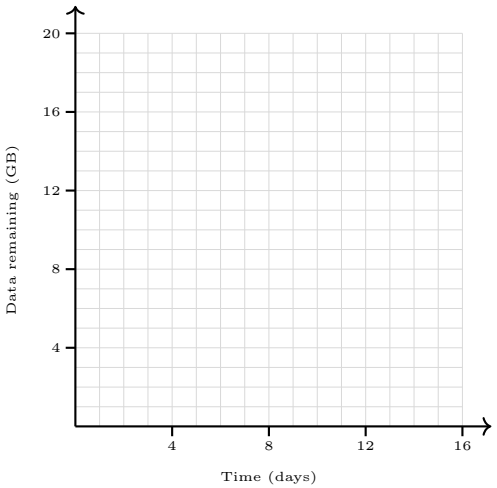
(d) $3x - 6y + 9 = 0$ and $x - 2y + 3 = 0$ are the same line.

(e) A line whose slope is 0 is a vertical line.

(f) In general form, A, B and C may never be negative.

Part F — Applications

15. Amira's monthly phone plan includes 20 GB of data. She uses data at a steady 1.25 GB per day. Let D be the data remaining (GB) after t days of the month.



(a) Write an equation for D in terms of t.

(b) Graph the relation on the axes provided.

(c) Convert your equation to general form. *(Clear the decimal first.)*

(d) State the D-intercept and the t-intercept, each with units, and explain what each one means for Amira.

(e) How much data is left after 6 days?

(f) Amira wants to still have 7.5 GB left when she starts streaming a series. On which day is that?

16. A print shop charges a one-time setup fee plus a fixed price per flyer. An order of 150 flyers cost \$86, and an order of 400 flyers cost \$161. Let C be the cost in dollars for n flyers.

(a) Determine the price per flyer and the setup fee, then write the equation for C in terms of n.

(b) Convert the equation to general form, with integer coefficients and a positive leading coefficient.

(c) Find the cost of 1000 flyers, then find the largest order possible on a budget of \$200.

17. A car-rental company offers two plans for a one-day rental. Plan A is a flat \$95 with unlimited distance. Plan B is \$35 plus \$0.25 per kilometre driven. Let C be the cost in dollars for d kilometres.

(a) Write an equation for each plan.

(b) State the slope of Plan A and explain what a slope of 0 means here. What kind of line is its graph?

(c) Determine the distance at which the two plans cost the same.

(d) Which plan is cheaper for a 180 km day? For a 400 km day? Show both costs each time.

(e) Write both equations in general form, using d and C. One of them will have a coefficient of 0 — explain why that is still a legitimate general form.

Part G — Challenge

18. **Find the missing coefficient.** Two lines have equations $3x - 4y + 7 = 0$ and $kx + 6y - 5 = 0$.

(a) State the slope of the first line, then write the slope of the second line as an expression in k.

(b) Determine the value of k that makes the two lines **perpendicular**, then verify that the two slopes multiply to -1.

(c) Determine the value of k that makes them **parallel**. Explain why that value means $kx + 6y - 5 = 0$ is no longer proper general form, and give the proper general form of that line.

19. **A triangle from three lines.** The three lines $y = 2x - 4$, $y = -x + 5$ and $y = -2$ enclose a triangle.

(a) Write all three equations in general form.

(b) Determine the coordinates of all three vertices. *(Each vertex is where two of the lines meet, so set the two expressions for y equal.)*

(c) Using the vertices, state the length of the horizontal side and the vertical height of the triangle, then find its area.

(d) Write the equation of the line through the top vertex that is perpendicular to $y = -2$, and explain why it cannot be written in the form $y = mx + b$.

20. **Building an equation from its intercepts and its points.**

(a) A line has equation $Ax + By - 12 = 0$. Its x-intercept is 4 and its y-intercept is -3. Determine A and B, write the equation in proper general form, and then convert it to slope-intercept form.

(b) A line has equation $6x + By + C = 0$ and passes through $(2, -1)$ and $(-3, 4)$. Determine B and C. *(Find the line's slope-intercept equation first, convert it to simplest general form, then scale the whole equation so the x-coefficient becomes 6.)*

Answer Key. 1. (a) From the table, x goes up by 2 while y goes up by 3, so $m = \frac{3}{2}$; $x = 0$ gives $b = -4$; $y = \frac{3}{2}x - 4$. (b) e.g. $(0, -4)$ and $(4, 2)$: $m = \frac{2-(-4)}{4-0} = \frac{6}{4} = \frac{3}{2}$, $b = -4$; $y = \frac{3}{2}x - 4$. (c) $m = \frac{5-(-7)}{6-(-2)} = \frac{12}{8} = \frac{3}{2}$; $-7 = \frac{3}{2}(-2) + b \Rightarrow b = -4$; $y = \frac{3}{2}x - 4$. (d) $m = \frac{3}{2}$; $2 = \frac{3}{2}(4) + b \Rightarrow b = -4$; $y = \frac{3}{2}x - 4$. (e) $y = \frac{3}{2}x - 4 \xrightarrow{\times 2} 2y = 3x - 8 \Rightarrow 3x - 2y - 8 = 0$. (f) $\frac{3}{2}(10) - 4 = 11$, so yes, $(10, 11)$ is on the line; $x = -8$ gives $y = -16$. **2.** (a) $-1 = -\frac{3}{4}(8) + b \Rightarrow b = 5$; $y = -\frac{3}{4}x + 5$, and $3x + 4y - 20 = 0$. (b) $m = \frac{-4-8}{5-(-3)} = -\frac{3}{2}$, $b = \frac{7}{2}$; $y = -\frac{3}{2}x + \frac{7}{2}$, and $3x + 2y - 7 = 0$. (c) Both points have $x = -5$, so it is the vertical line $x = -5$, i.e. $x + 5 = 0$. (d) Both points have $y = -3$, so it is the horizontal line $y = -3$, i.e. $y + 3 = 0$ (here $A = 0$). (e) Part (c): a vertical line has an undefined slope, so there is no m to put into $y = mx + b$. Part (d) *does* fit, because a horizontal line has slope 0: $y = 0x - 3$. **3.** (a) $-6y = -4x - 9 \Rightarrow y = \frac{2}{3}x + \frac{3}{2}$; slope $\frac{2}{3}$. (b) $m = \frac{2}{3}$: $5 = \frac{2}{3}(-6) + b \Rightarrow b = 9$; $y = \frac{2}{3}x + 9$, so $2x - 3y + 27 = 0$. (c) $m = -\frac{3}{2}$: $5 = -\frac{3}{2}(-6) + b \Rightarrow b = -4$; $y = -\frac{3}{2}x - 4$, so $3x + 2y + 8 = 0$. **4.** (a) $5a - 14 - 16 = 0 \Rightarrow a = 6$. (b) $m = \frac{14-5}{8-2} = \frac{3}{2}$, $b = 2$, so $y = \frac{3}{2}x + 2$ and $k = \frac{3}{2}(-4) + 2 = -4$. (c) $2 = 3m - 7 \Rightarrow m = 3$. (d) $6 - 3B - 12 = 0 \Rightarrow B = -2$. **5.** (a) Every step of $+3$ in x gives -4 in y, so $m = -\frac{4}{3}$ throughout; $x = 0$ gives $b = 5$, so $y = -\frac{4}{3}x + 5$. (b) $\times 3$: $3y = -4x + 15 \Rightarrow 4x + 3y - 15 = 0$. (c) $y = -\frac{4}{3}(12) + 5 = -11$. (d) $25 = -\frac{4}{3}x + 5 \Rightarrow -\frac{4}{3}x = 20 \Rightarrow x = -15$. **6.** (a) $5x + 6y - 4 = 0$ (b) $7x - 4y - 12 = 0$ (c) $2x + 6y + 15 = 0$ (d) $2x - 9 = 0$. **7.** (a) $y = \frac{3}{4}x + \frac{2}{3}$; $m = \frac{3}{4}$, y-int $\left(0, \frac{2}{3}\right)$, x-int $\left(-\frac{8}{9}, 0\right)$. (b) $y = -\frac{2}{5}x + \frac{4}{3}$; $m = -\frac{2}{5}$, y-int $\left(0, \frac{4}{3}\right)$, x-int $\left(\frac{10}{3}, 0\right)$. **8.** (a) (i), (ii), (iii) and (iv) are all the same line: (i)$\times 5$ gives $2x - 5y - 15 = 0$; (iii)$\times(-1)$ gives $2x - 5y - 15 = 0$; (iv)$\div 2$ gives $2x - 5y - 15 = 0$. (v) is different. (b) (ii) is proper. (iii) has $A = -2$, which is negative. (iv) has a common factor of 2 in all three coefficients, so it is not reduced. (c) (v) has $m = \frac{5}{2}$ and $b = -3$; it shares the y-intercept -3 with the others but its slope is $\frac{5}{2}$, not $\frac{2}{5}$, so it is a different line that merely crosses the others at $(0, -3)$. **9.** (a) e.g. $(-3, 6)$ and $(3, -2)$ (also $(0, 2)$, $(-6, 10)$, $(6, -6)$). (b) $m = \frac{-2-6}{3-(-3)} = \frac{-8}{6} = -\frac{4}{3}$. (c) $b = 2$, so $y = -\frac{4}{3}x + 2$. (d) $\times 3$: $3y = -4x + 6 \Rightarrow 4x + 3y - 6 = 0$. (e) $x = 12$: $y = -16 + 2 = -14$; $-30 = -\frac{4}{3}x + 2 \Rightarrow -\frac{4}{3}x = -32 \Rightarrow x = 24$. **10.** (a) (i) $m = \frac{2}{3}$, $b = -3$. (ii) $2y = -x + 8 \Rightarrow y = -\frac{1}{2}x + 4$, so $m = -\frac{1}{2}$, $b = 4$. (iii) $y = -2$: $m = 0$, $b = -2$. (iv) $x = -5$: no slope-intercept form. (b) $x = -5$ is vertical, so its slope is undefined and it can never be written $y = mx + b$; $y = -2$ is horizontal with slope 0, which is a perfectly ordinary number, so it *can* be written $y = 0x - 2$. (c) Lines (i) and (ii) meet at $(6, 1)$: $\frac{2}{3}(6) - 3 = 1$ and $6 + 2(1) - 8 = 0$. **11.** (a) $(2, 1)$. (b) L_1 passes through $(0, -1)$ and $(1, 0)$, so $m = 1$, $b = -1$, $y = x - 1$, giving $x - y - 1 = 0$. (c) L_2 passes through $(0, 5)$ and $(1, 3)$, so $m = -2$, $b = 5$, $y = -2x + 5$, giving $2x + y - 5 = 0$. (d) $2 - 1 - 1 = 0$ ✓ and $2(2) + 1 - 5 = 0$ ✓. **12.** (a) Step 3 is wrong: $-2(-4) = +8$, not -8. (b) $9 = 8 + b \Rightarrow b = 1$, so $y = -2x + 1$ and the general form is $2x + y - 1 = 0$. (c) $(-4, 9)$: $2(-4) + 9 - 1 = 0$ ✓; $(2, -3)$: $2(2) - 3 - 1 = 0$ ✓. **13.** (a) Step 1: he multiplied only the first two terms by 5 and left the $+2$ alone — every term must be multiplied, so the right side is $-3x + 10$. (b) $5y = -3x + 10 \Rightarrow 3x + 5y - 10 = 0$. (c) Malik's: $3(0) + 5(2) - 2 = 8 \neq 0$, so $(0, 2)$ is not on his line; correct: $3(0) + 5(2) - 10 = 0$ ✓. **14.** (a) False — vertical lines such as $x = 4$ have no slope, so they cannot. (b) True — $y = mx + b$ becomes $mx - y + b = 0$, and $x = k$ becomes $x - k = 0$. (c) False — parallel lines have equal *slopes*; two lines sharing a y-intercept simply cross there (e.g. $y = 2x + 1$ and $y = 5x + 1$). (d) True — dividing $3x - 6y + 9 = 0$ by 3 gives $x - 2y + 3 = 0$, and both are $y = \frac{1}{2}x + \frac{3}{2}$. (e) False — slope 0 is a *horizontal* line; a vertical line has an undefined slope. (f) False — only A is made positive by convention; B and C may be negative, as in $3x - 4y - 12 = 0$. **15.** (a) $D = -1.25t + 20$. (b) A straight line from $(0, 20)$ to $(16, 0)$, passing through $(8, 10)$. (c) $\times 4$: $4D = -5t + 80 \Rightarrow 5t + 4D - 80 = 0$. (d) D-intercept 20 GB — the data she starts the month with; t-intercept 16 days — the day her data runs out. (e) $D = -1.25(6) + 20 = 12.5$ GB. (f) $7.5 = -1.25t + 20 \Rightarrow 1.25t = 12.5 \Rightarrow t = 10$ days. **16.** (a) $m = \frac{161-86}{400-150} = \frac{75}{250} = 0.30$ dollars per flyer; $86 = 0.30(150) + b \Rightarrow b = 41$, so the setup fee is \$41 and $C = 0.3n + 41$. (b) $\times 10$: $10C = 3n + 410 \Rightarrow 3n - 10C + 410 = 0$. (c) $C = 0.3(1000) + 41 = \$341$; $200 = 0.3n + 41 \Rightarrow 0.3n = 159 \Rightarrow n = 530$ flyers. **17.** (a) Plan A: $C = 95$; Plan B: $C = 0.25d + 35$. (b) Plan A has slope 0 — the cost does not change at all as d increases; its graph is a horizontal line. (c) $95 = 0.25d + 35 \Rightarrow 0.25d = 60 \Rightarrow d = 240$ km. (d) 180 km: A = \$95, B = $0.25(180) + 35 = \$80$, so B is cheaper; 400 km: A = \$95, B = $0.25(400) + 35 = \$135$, so A is cheaper. (e) Plan A: $C - 95 = 0$ (that is $0d + 1C - 95 = 0$); Plan B: $\times 4$ gives $4C = d + 140 \Rightarrow d - 4C + 140 = 0$. A coefficient of 0 is allowed because general form only requires A, B, C to be integers with not both A and B zero — a missing variable simply means the line does not depend on it. **18.** (a) $3x - 4y + 7 = 0 \Rightarrow y = \frac{3}{4}x + \frac{7}{4}$, so $m_1 = \frac{3}{4}$; $kx + 6y - 5 = 0 \Rightarrow y = -\frac{k}{6}x + \frac{5}{6}$, so $m_2 = -\frac{k}{6}$. (b) $\frac{3}{4} \cdot \left(-\frac{k}{6}\right) = -1 \Rightarrow -\frac{k}{8} = -1 \Rightarrow k = 8$; then $m_2 = -\frac{8}{6} = -\frac{4}{3}$ and $\frac{3}{4} \cdot \left(-\frac{4}{3}\right) = -1$ ✓. (c) $-\frac{k}{6} = \frac{3}{4} \Rightarrow k = -\frac{9}{2}$. That is not an integer, so $-\frac{9}{2}x + 6y - 5 = 0$ breaks the integer requirement (and A is negative); multiplying through by -2 gives $9x - 12y + 10 = 0$, which checks out since $12y = 9x + 10 \Rightarrow y = \frac{3}{4}x + \frac{5}{6}$. **19.** (a) $2x - y - 4 = 0$, $x + y - 5 = 0$, $y + 2 = 0$. (b) $2x - 4 = -x + 5 \Rightarrow x = 3$, $y = 2$, so $(3, 2)$; $2x - 4 = -2 \Rightarrow x = 1$, so $(1, -2)$; $-x + 5 = -2 \Rightarrow x = 7$, so $(7, -2)$. (c) Horizontal side from $(1, -2)$ to $(7, -2)$ has length 6; the height from $y = -2$ up to $y = 2$ is 4; area $= \frac{1}{2}(6)(4) = 12$ square units. (d) A line perpendicular to the horizontal line $y = -2$ is vertical, so through $(3, 2)$ it is $x = 3$, or $x - 3 = 0$; a vertical line has an undefined slope, so there is no m for $y = mx + b$. **20.** (a) x-intercept 4: $4A - 12 = 0 \Rightarrow A = 3$; y-intercept -3: $-3B - 12 = 0 \Rightarrow B = -4$. So $3x - 4y - 12 = 0$, and $4y = 3x - 12 \Rightarrow y = \frac{3}{4}x - 3$. Check: $3(4) - 4(0) - 12 = 0$ and $3(0) - 4(-3) - 12 = 0$ ✓. (b) $m = \frac{4-(-1)}{-3-2} = \frac{5}{-5} = -1$; $-1 = -(2) + b \Rightarrow b = 1$, so $y = -x + 1$ and the simplest general form is $x + y - 1 = 0$. Multiplying by 6 gives $6x + 6y - 6 = 0$, so $B = 6$ and $C = -6$. Check $(2, -1)$: $12 - 6 - 6 = 0$ ✓.